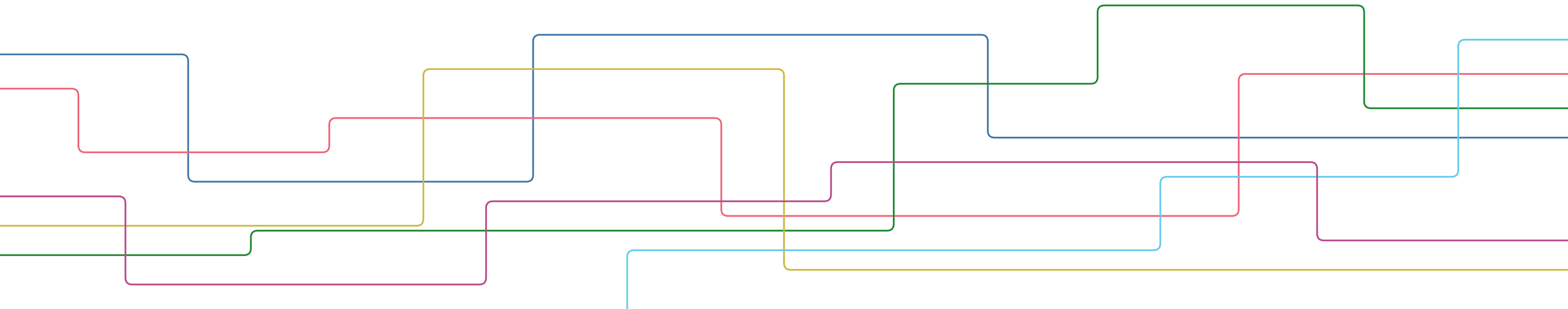
Programming for Data Science

– Gaussian Processes

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Outline

Gaussian distribution

- Covariance
- Sampling random vectors
- Conditional Gaussian distributions

Gaussian process

- Fundamentals and computation
- Gaussian process regression
- Kernel functions

Gaussian distribution

- A random vector $\mathbf{x} = [x_1, x_2, \ldots, x_D]^T$ is said to have a multivariate Gaussian distribution if all linear combinations of the vector $\mathbf{x}$ are Gaussian-distributed:

$$\mathbf{a}^T\mathbf{x} = a_1x_1 + a_2x_2 + \cdots + a_Dx_D \sim \mathcal{N}(m, V)$$

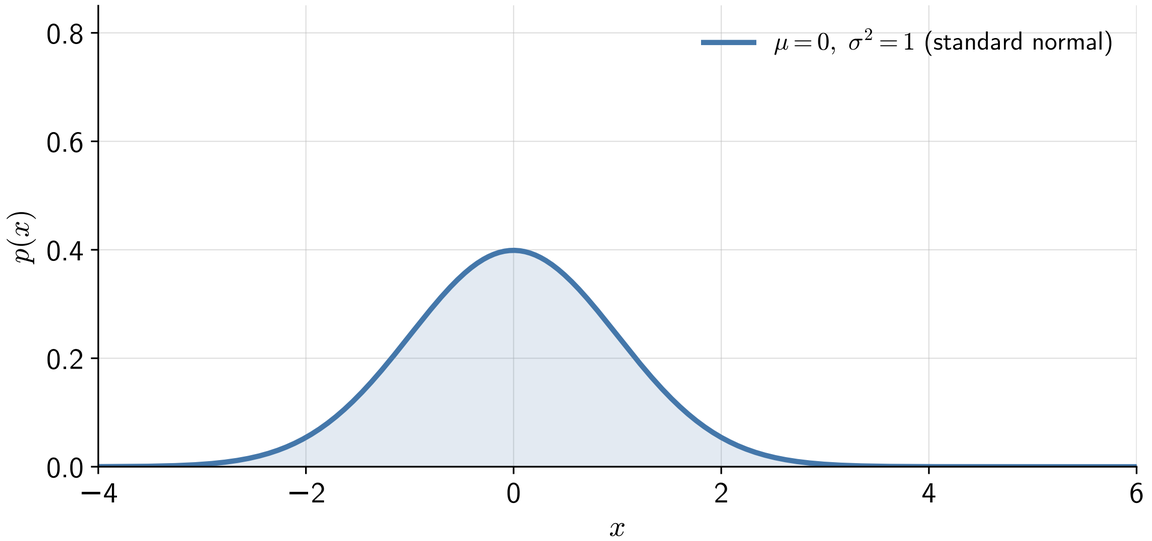
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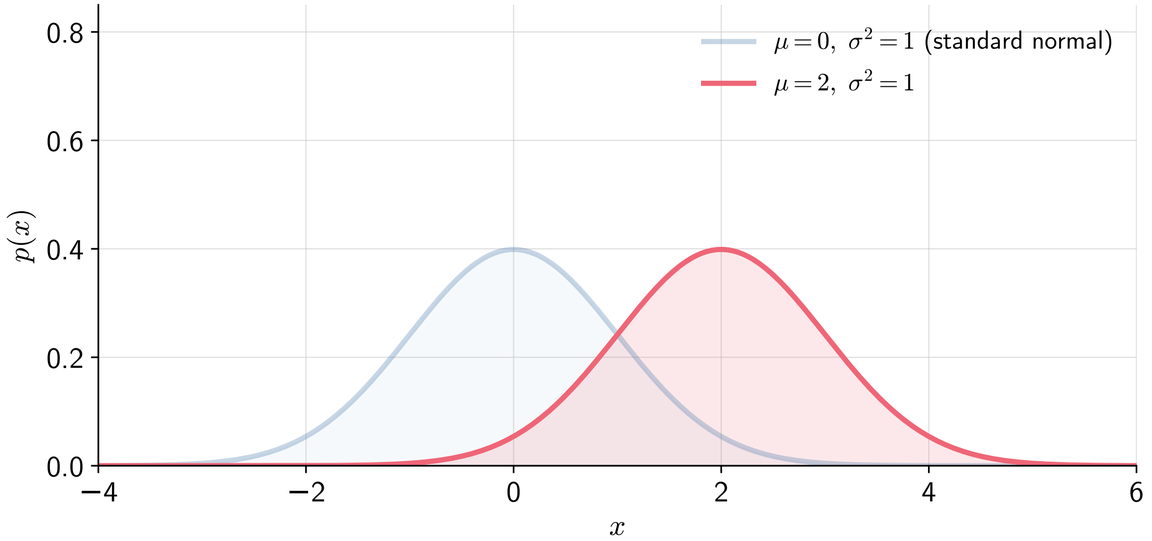


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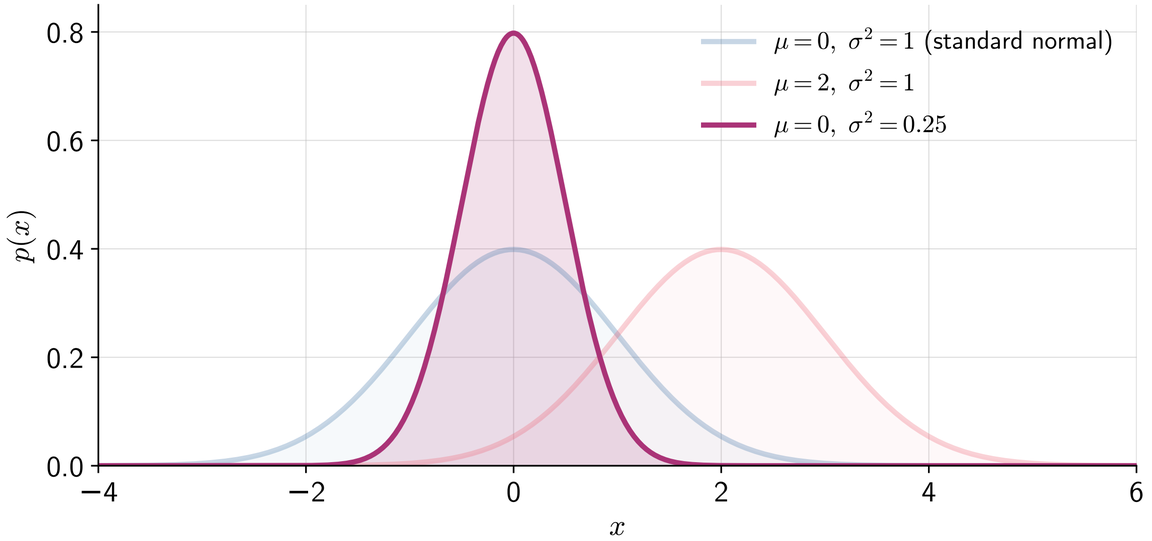


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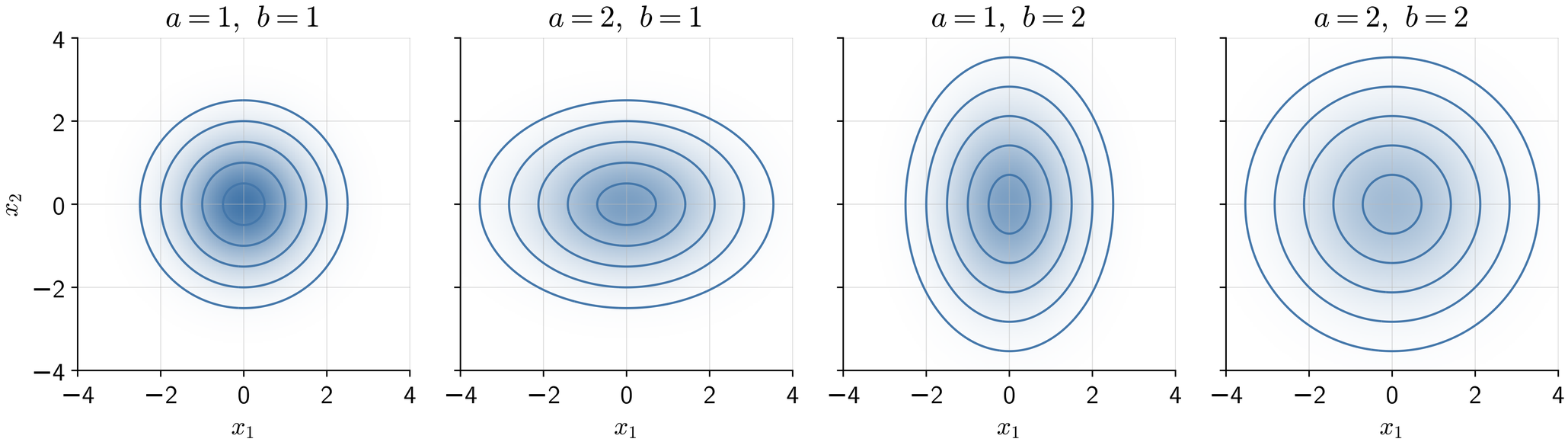
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Interpretation of the covariance matrix

- The diagonal of the covariance matrix controls the scaling/marginal variances

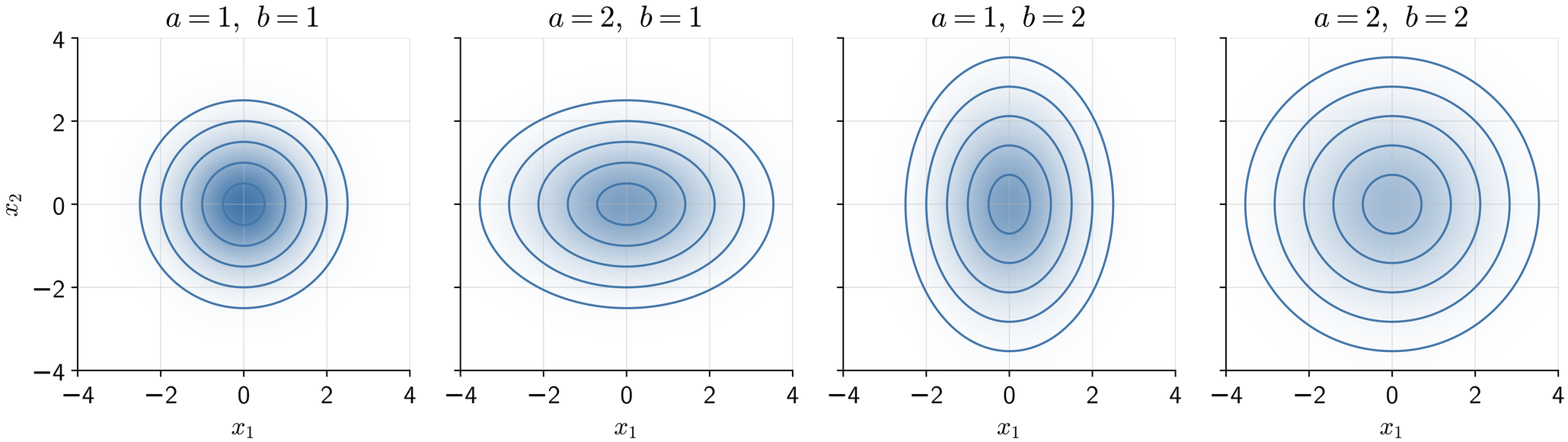
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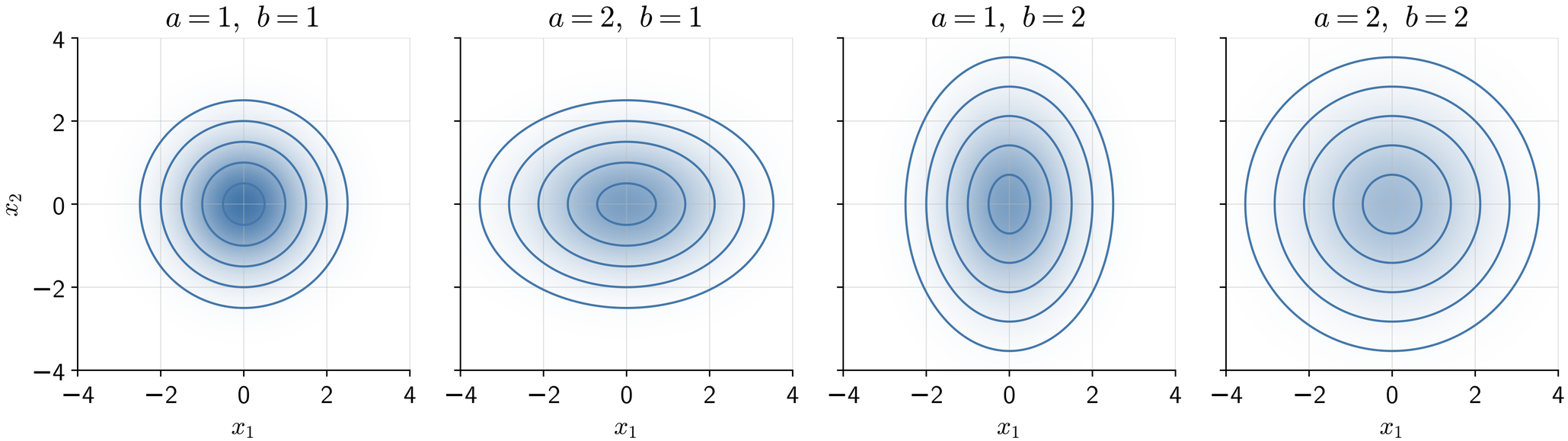


- If Σ is diagonal, then x_1 and x_2 are uncorrelated? True or false?

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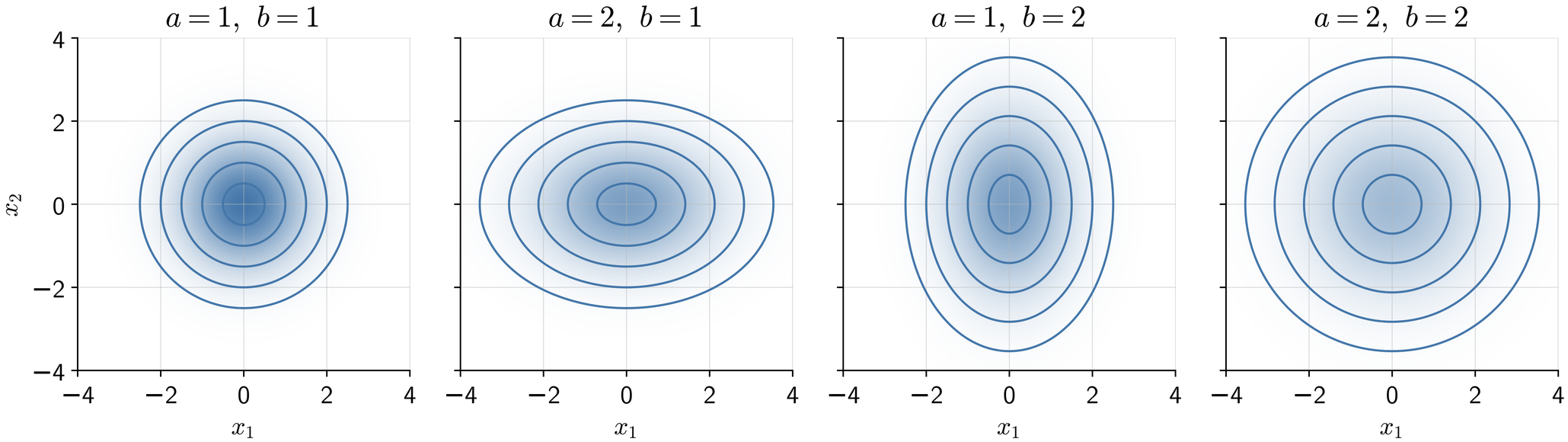


- What is the volume (integral) of the density?

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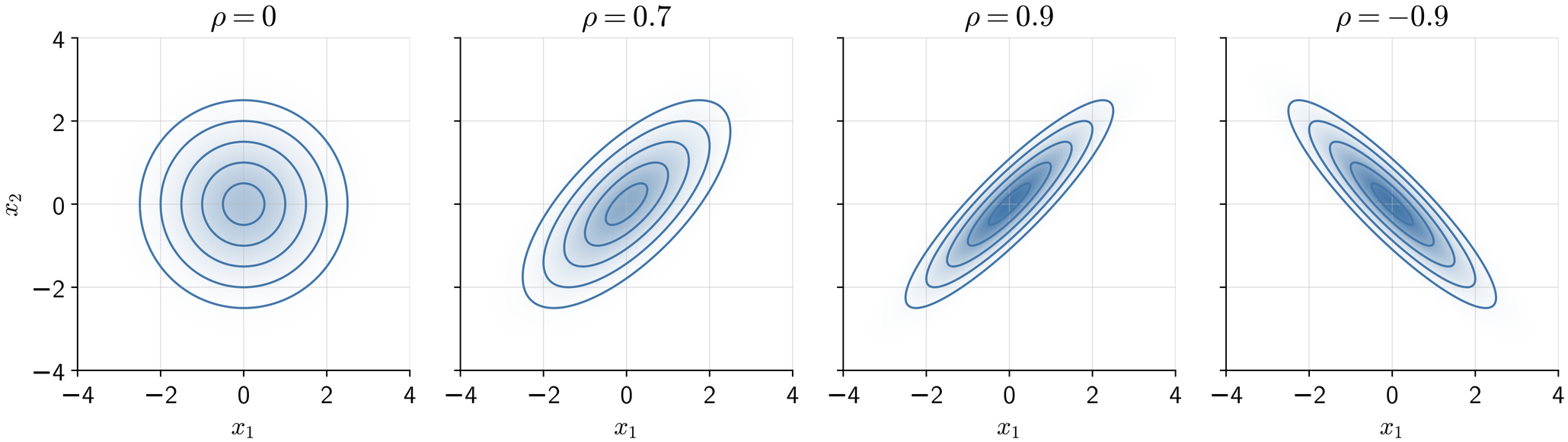


- Which of the four densities has the highest peak?

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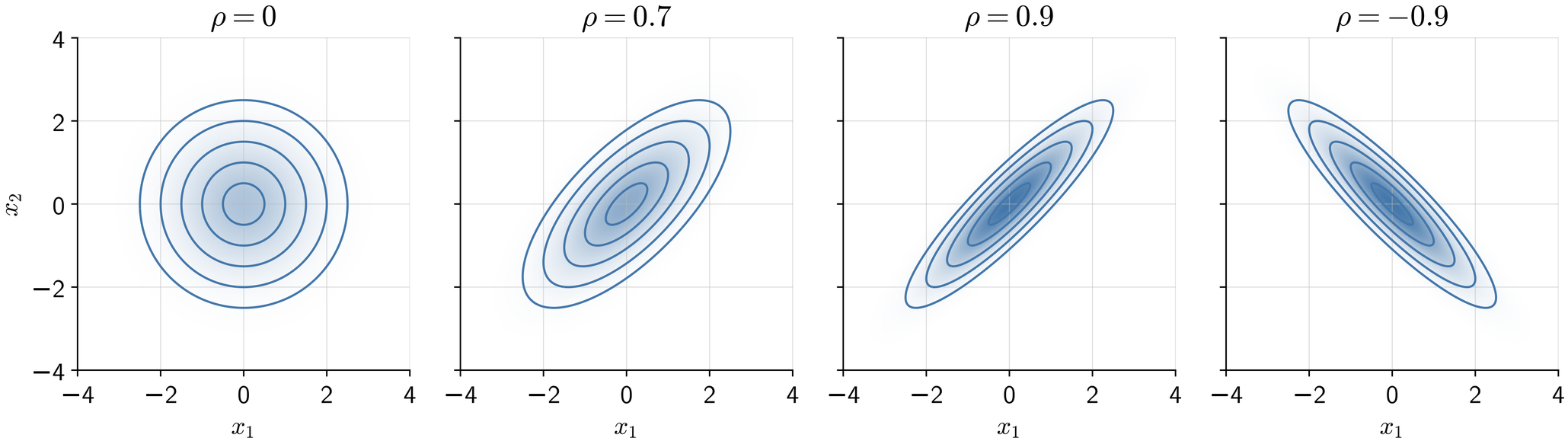
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Inverse of whitening operation

Sampling from a multivariate Gaussian

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Gaussian samples

```python
import numpy as np

x = np.random.randn(100, 2)     # sample 100 observations (D = 2)

m = [-2.0, 1.0]                 # mean vector of y
V = [[1.0, 0.7], [0.7, 2.0]]    # covariance matrix of y

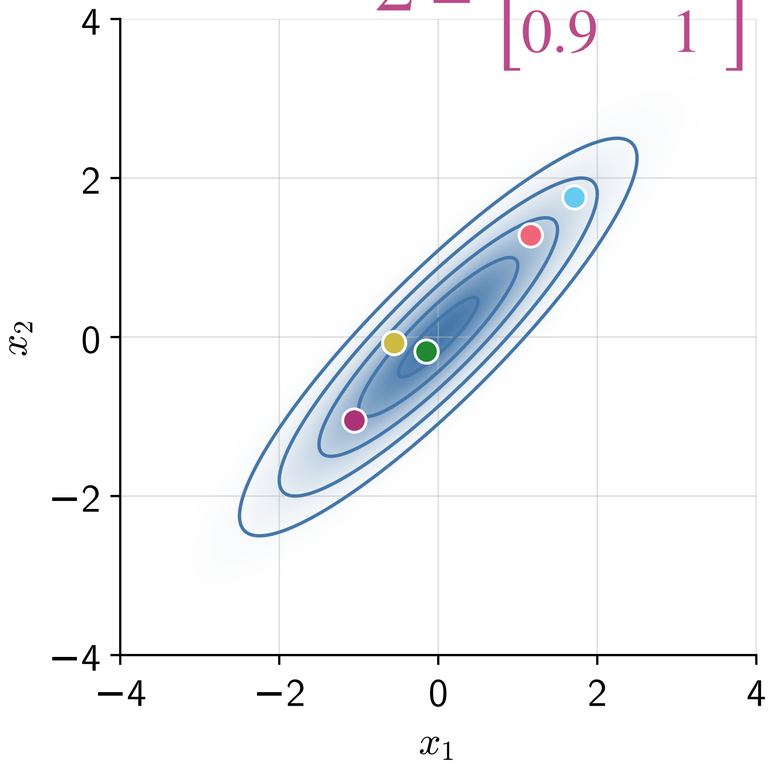
L = np.linalg.cholesky(V)       # matrix square root (Cholesky factor)
y = (x @ L.T) + m               # transform x samples into y

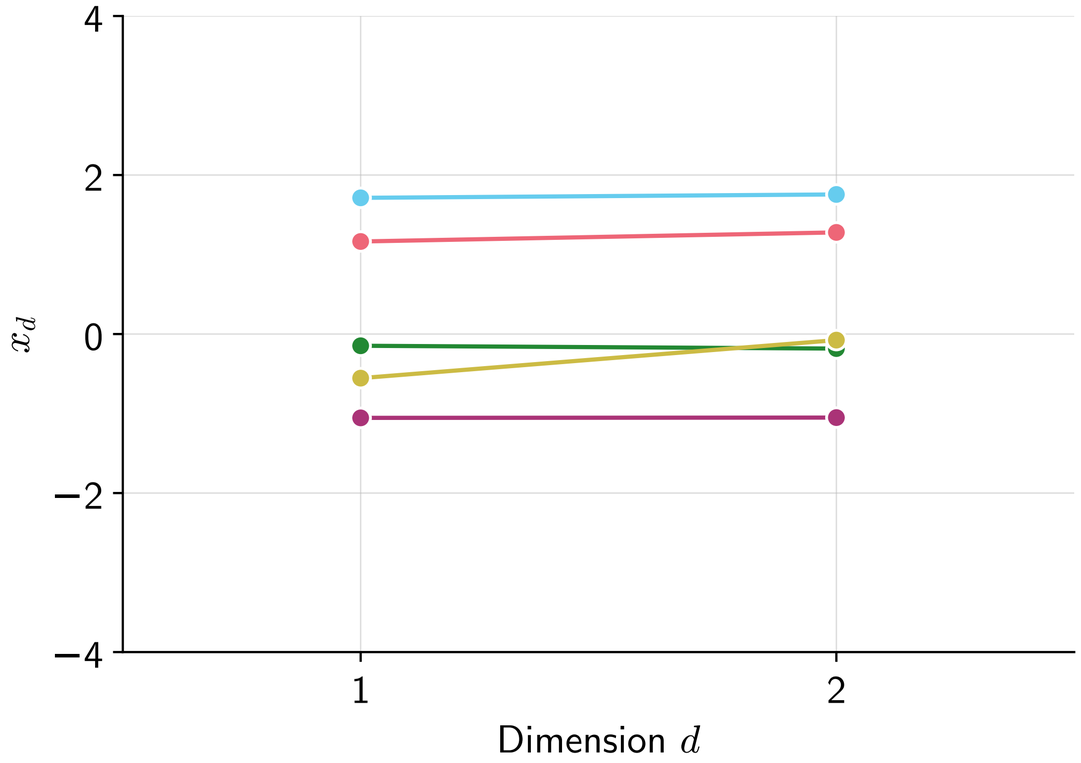
```

Samples in higher dimensions

- Samples in $D = 2$

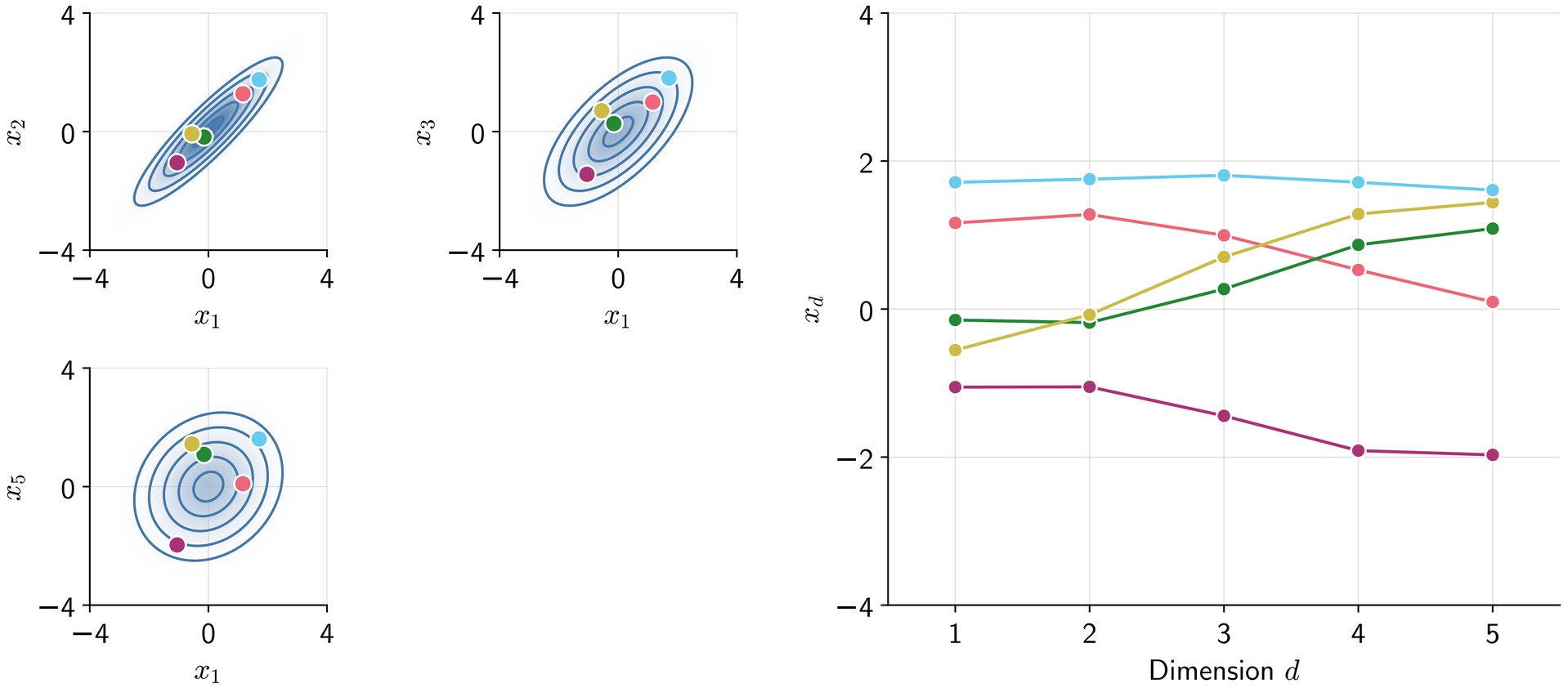
$$\Sigma = \begin{bmatrix} 1 & 0.9 \\ 0.9 & 1 \end{bmatrix}$$





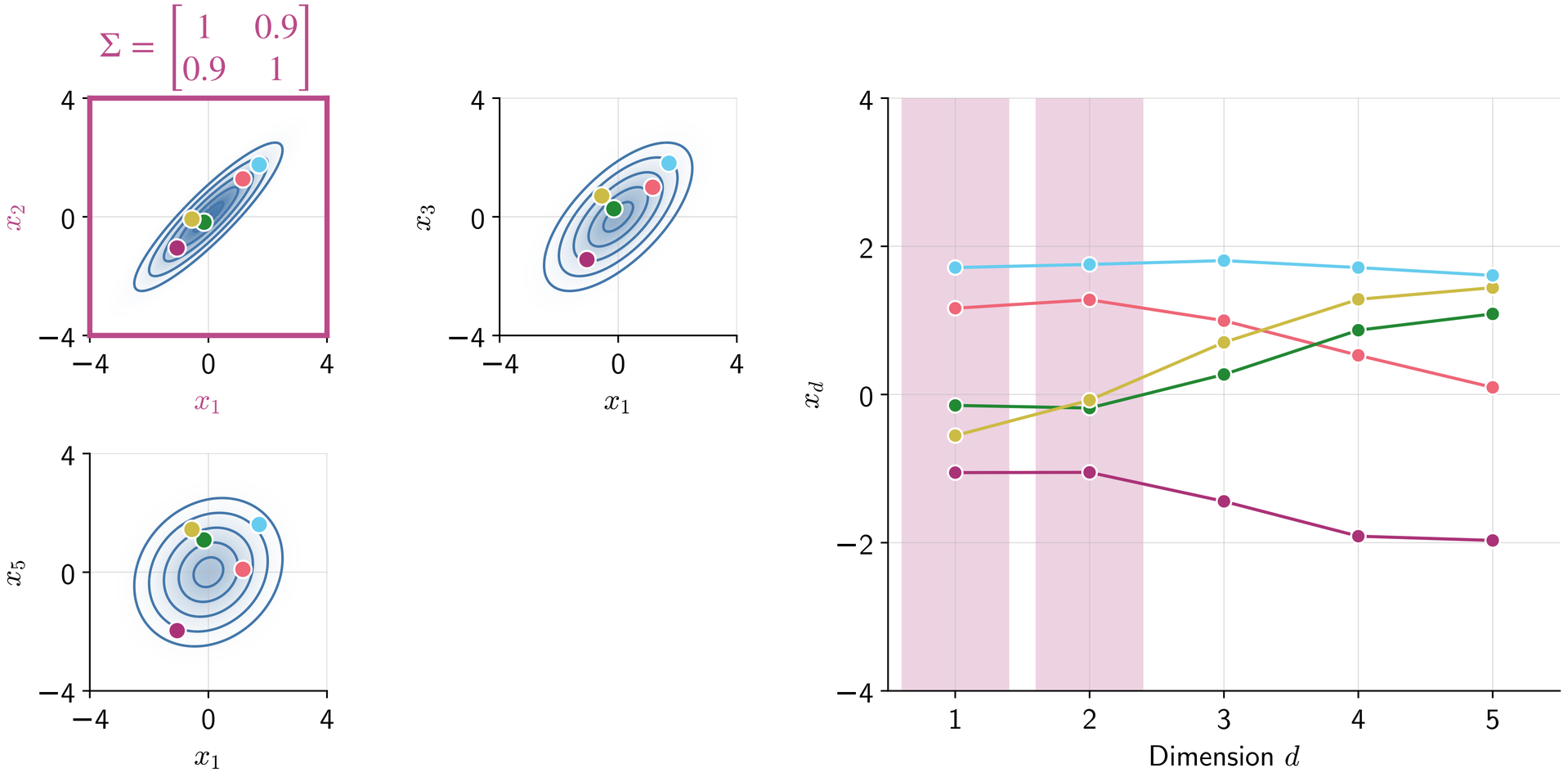
Samples in higher dimensions

- Samples in $D = 5$



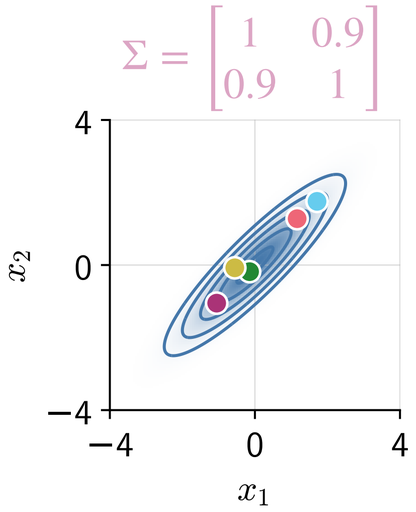
Samples in higher dimensions

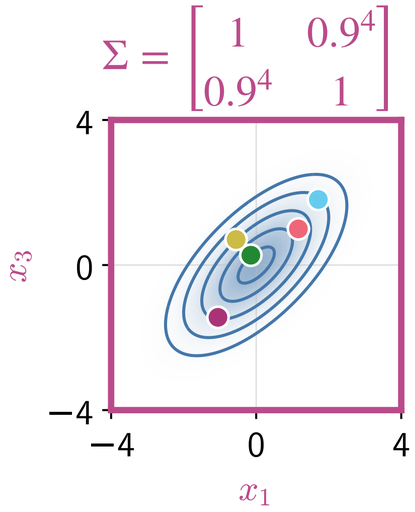
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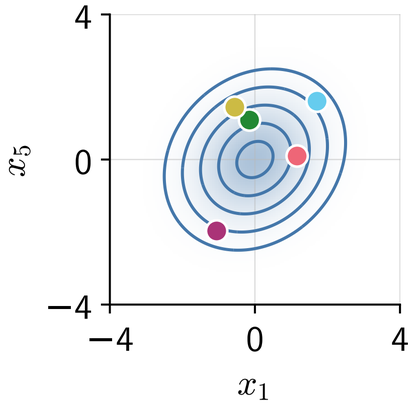


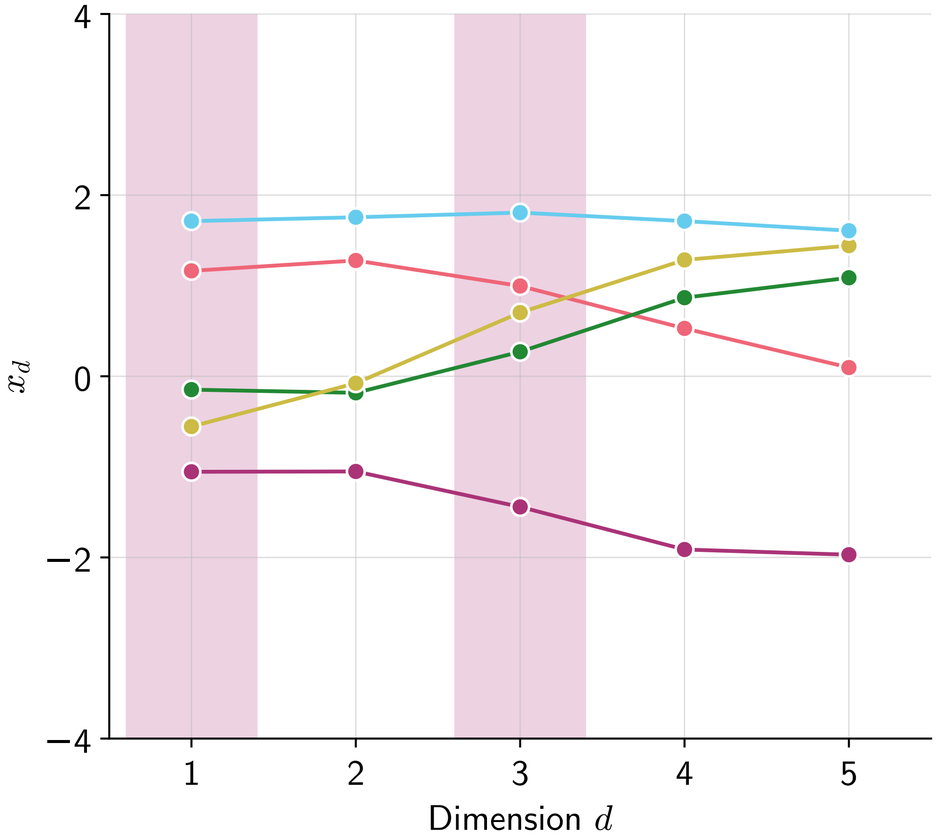
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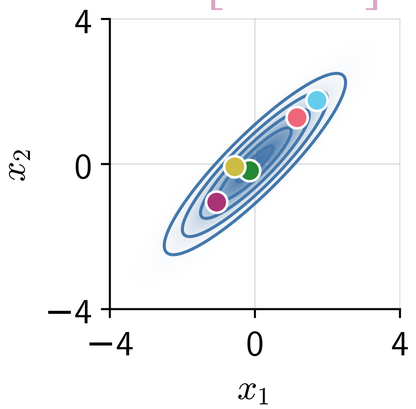




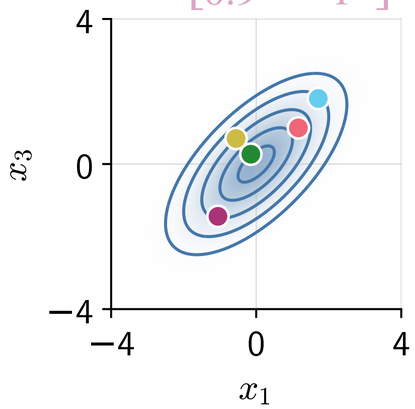
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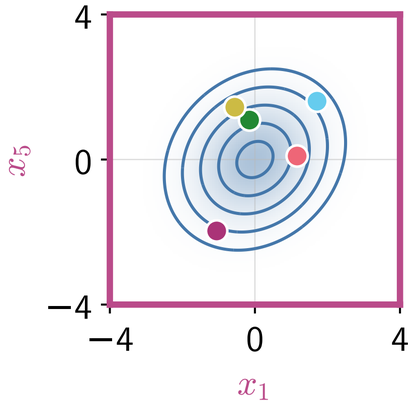
- Samples in $D = 5$

$$\Sigma = \begin{bmatrix} 1 & 0.9 \\ 0.9 & 1 \end{bmatrix}$$

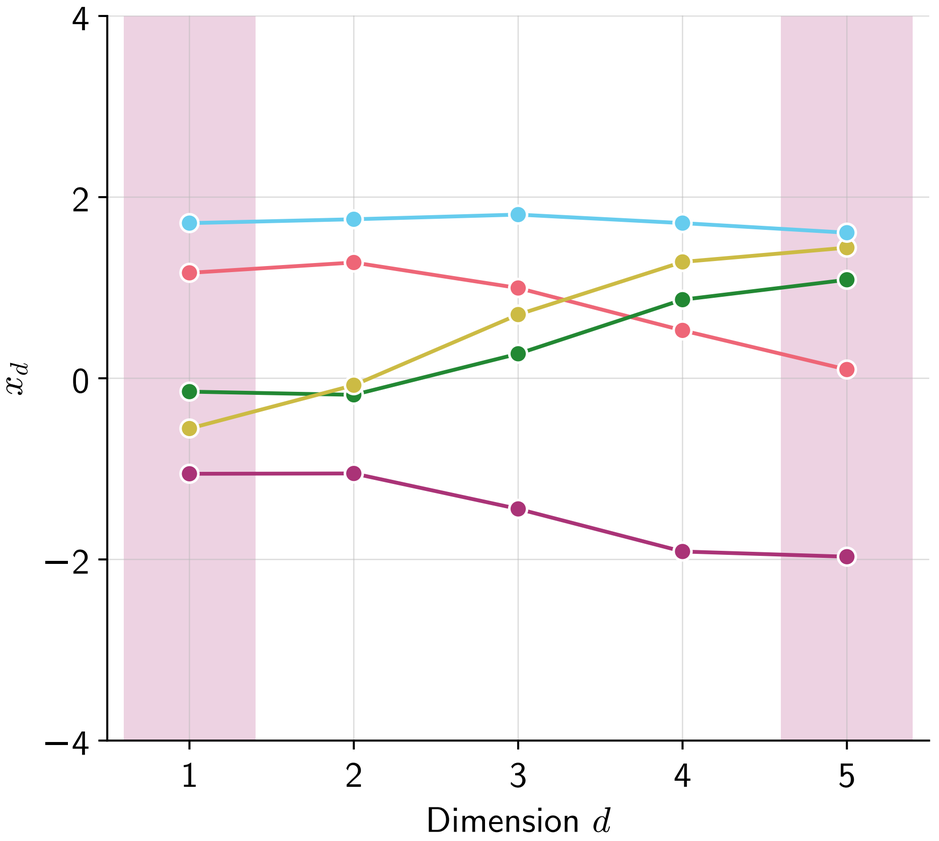


$$\Sigma = \begin{bmatrix} 1 & 0.9^4 \\ 0.9^4 & 1 \end{bmatrix}$$



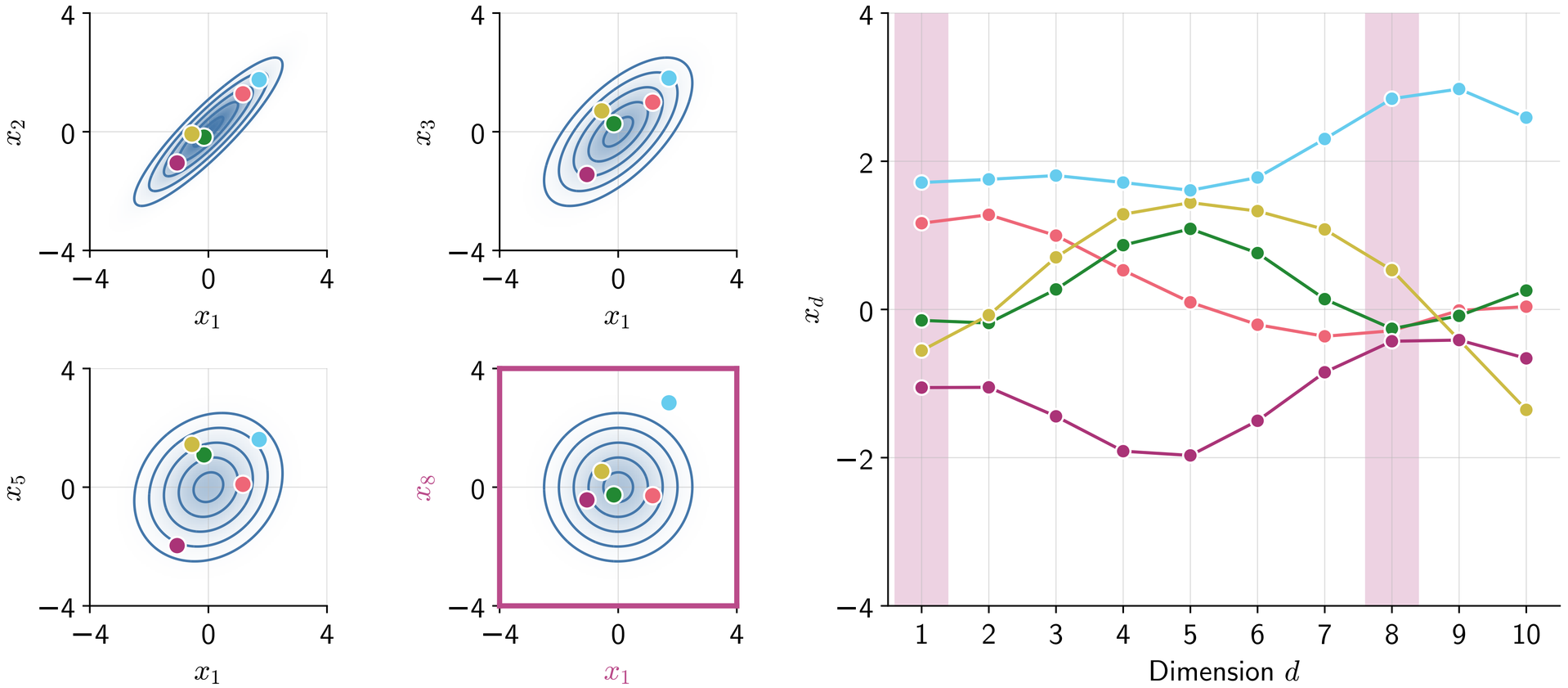


$$\Sigma = \begin{bmatrix} 1 & 0.9^{16} \\ 0.9^{16} & 1 \end{bmatrix}$$



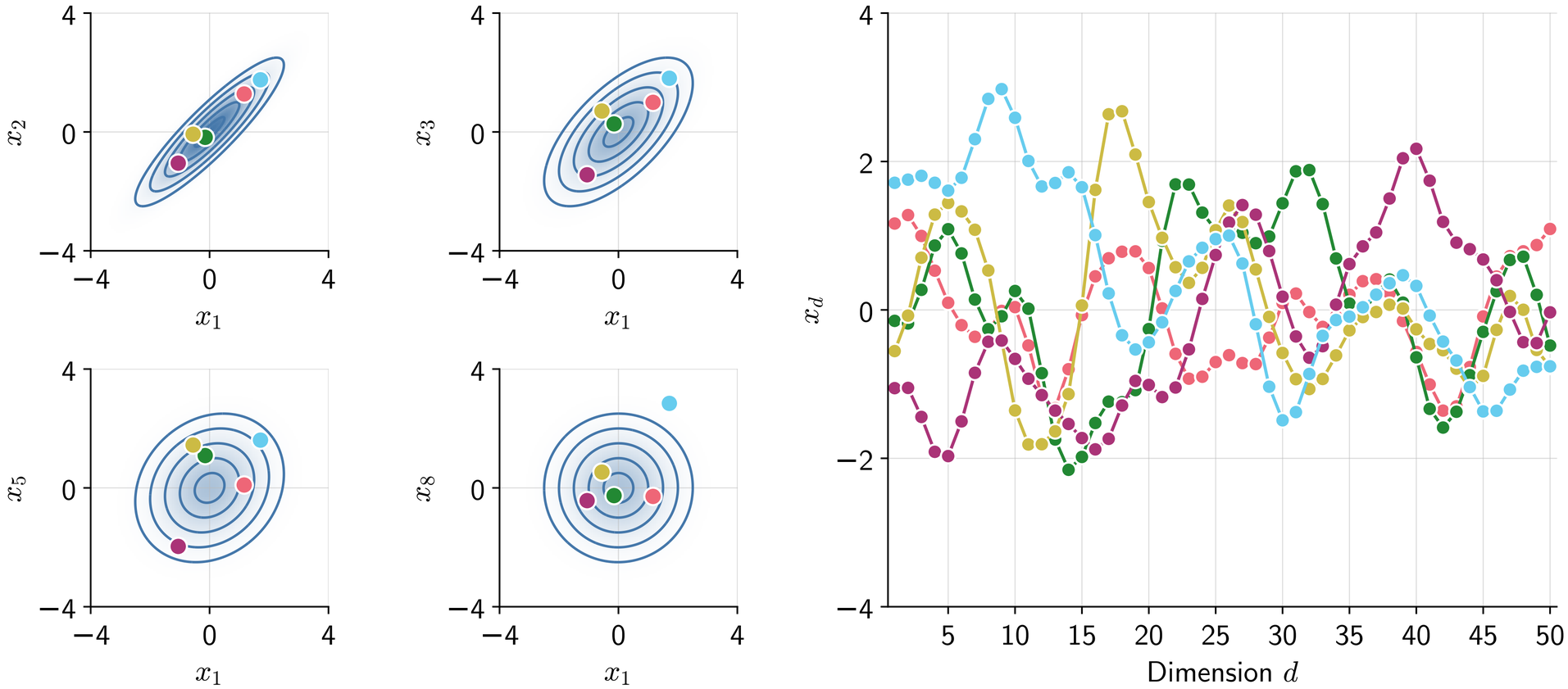
Samples in higher dimensions

- Samples in $D = 10$



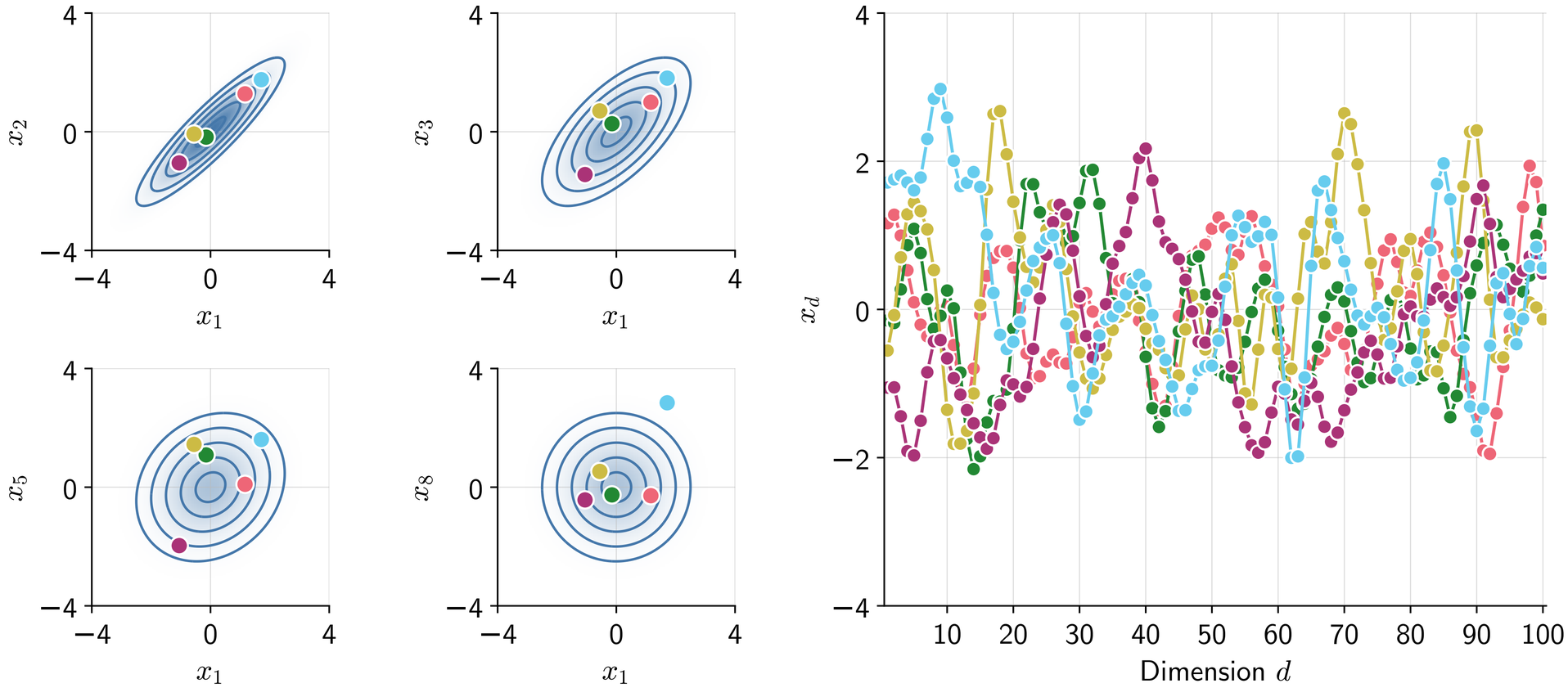
Samples in higher dimensions

- Samples in $D = 50$



Samples in higher dimensions

- Samples in $D = 100$



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- *We say, $p(y \mid x)$ is the conditional probability of y given (conditioned on) x*

Conditioning a multivariate Gaussian

$$p(y \mid x) = \frac{p(x, y)}{p(x)}$$

- Let $\mathbf{x}_1$ and $\mathbf{x}_2$ be jointly Gaussian distributed:

$$p(\mathbf{x}_1, \mathbf{x}_2) = \mathcal{N}\left(\begin{bmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \end{bmatrix} \mid \begin{bmatrix} \mathbf{m}_1 \\ \mathbf{m}_2 \end{bmatrix}, \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix} \right)$$

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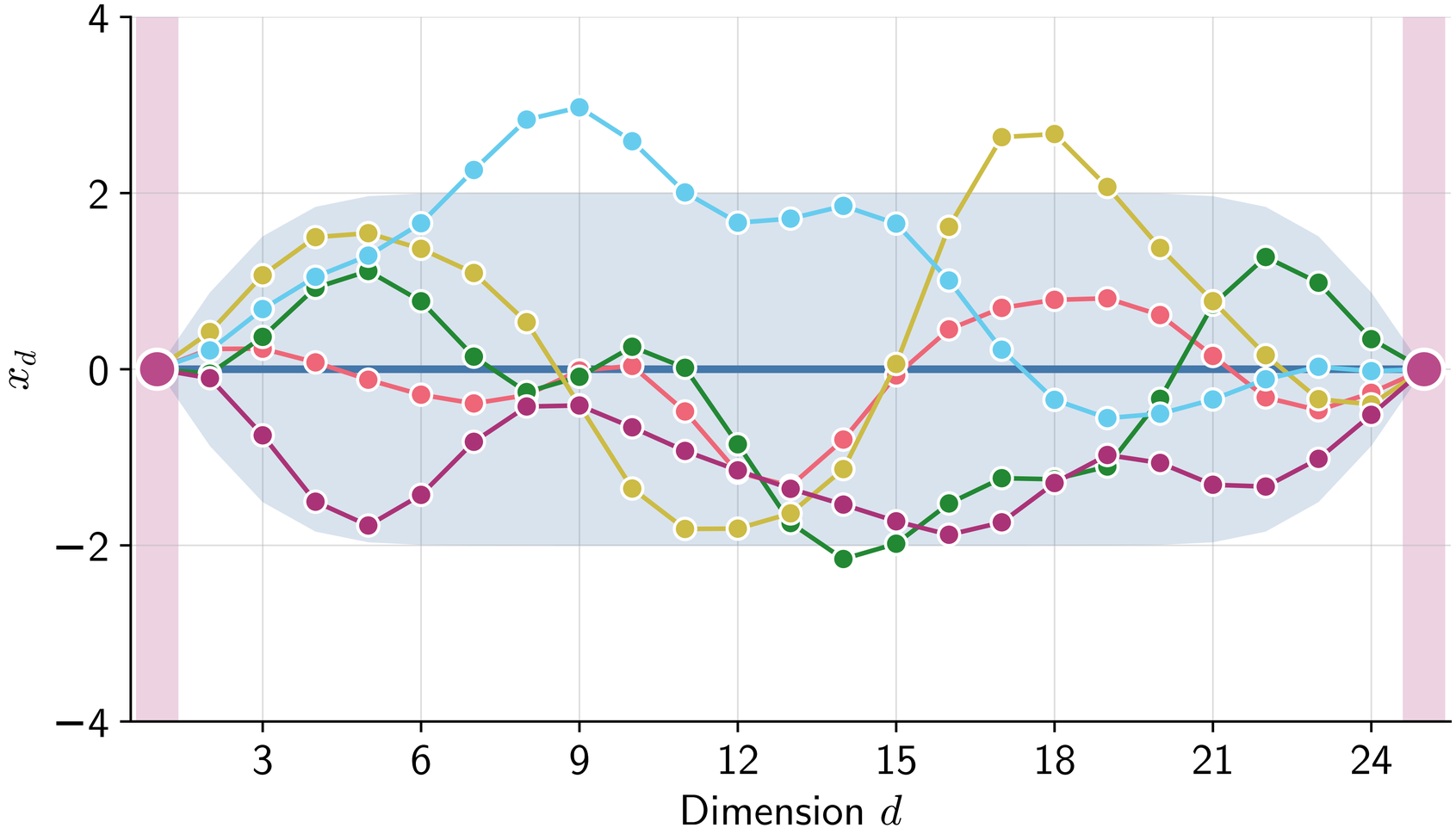
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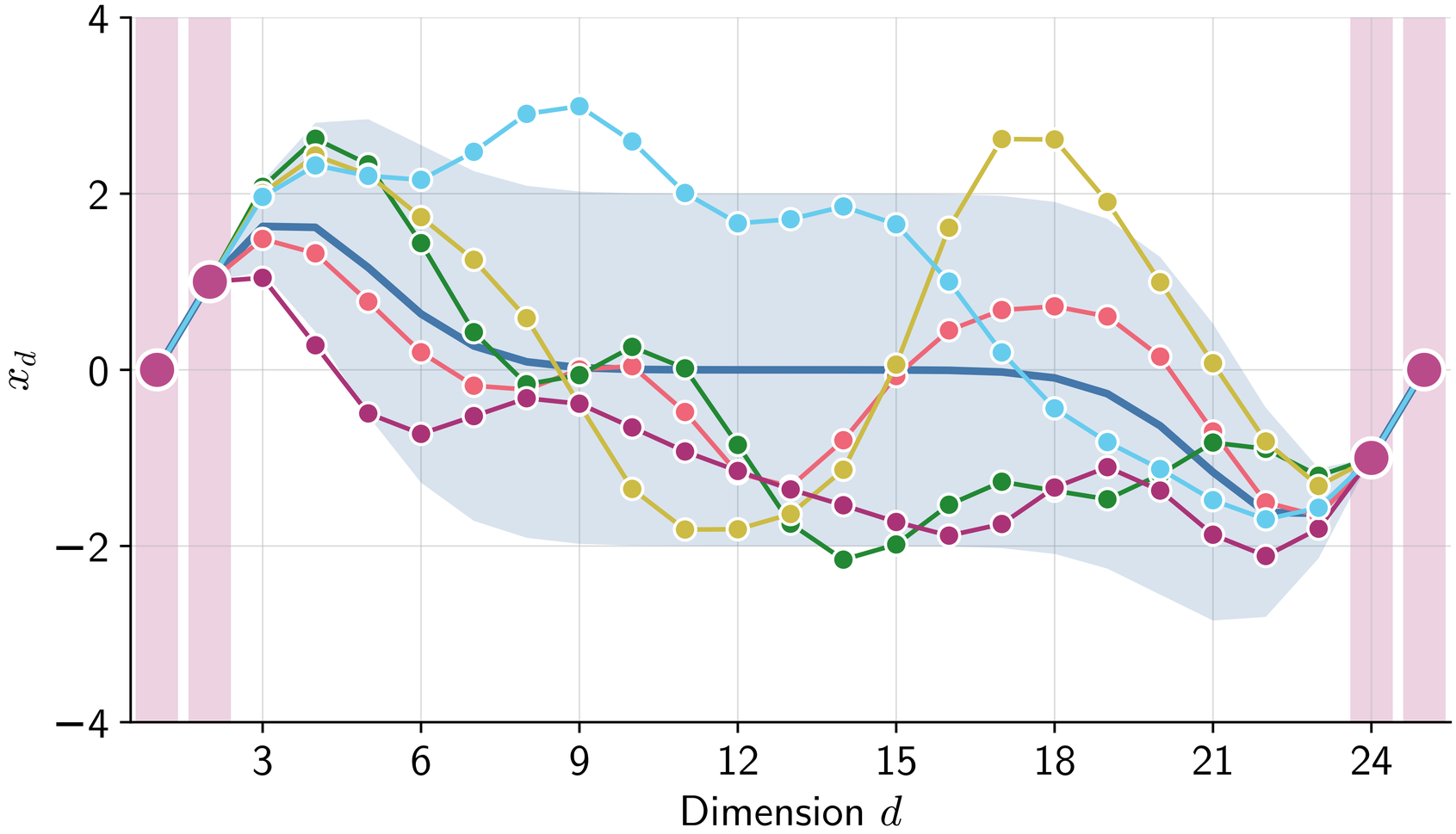
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mean vector covariance matrix

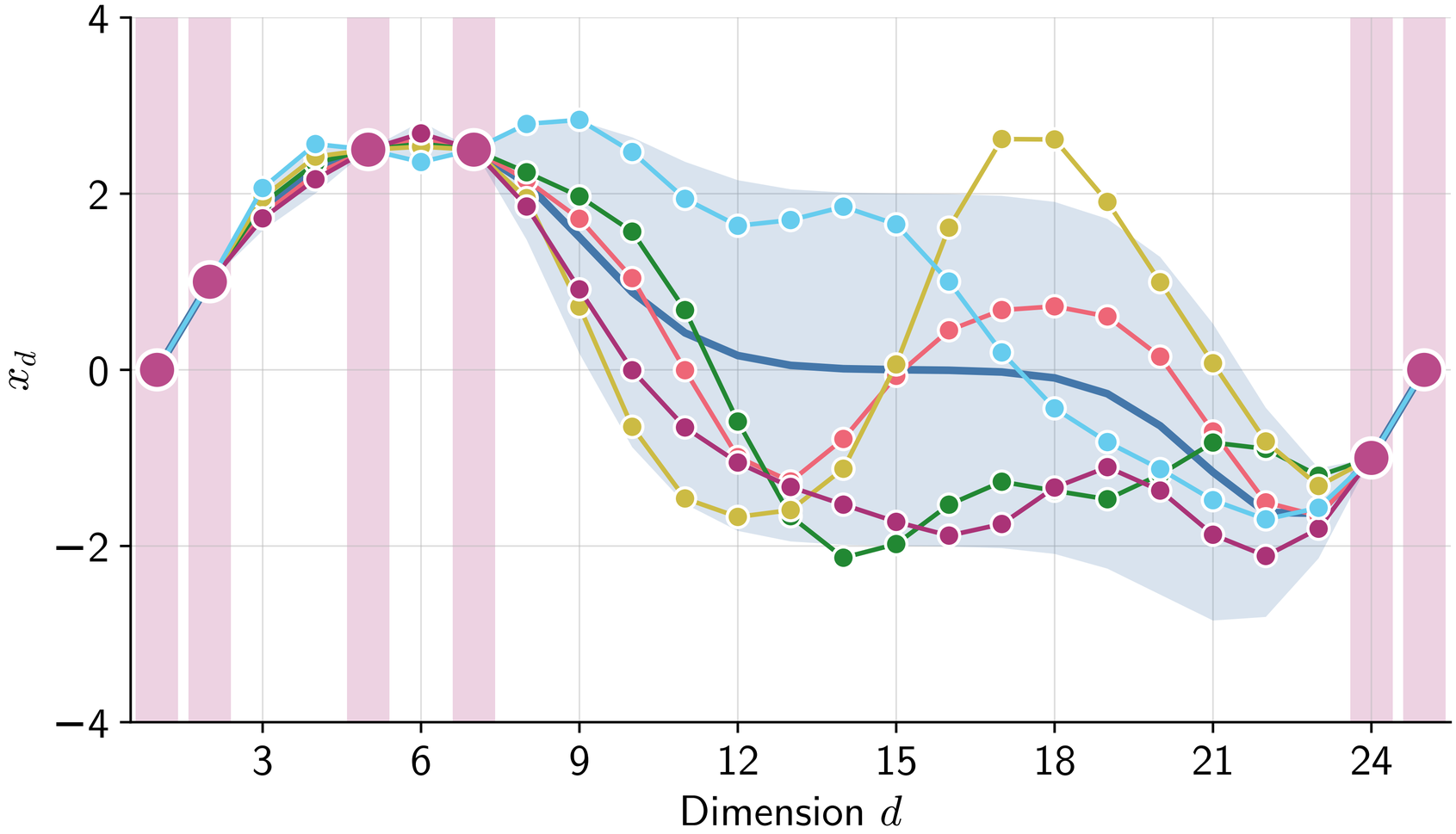
Conditioning a multivariate Gaussian



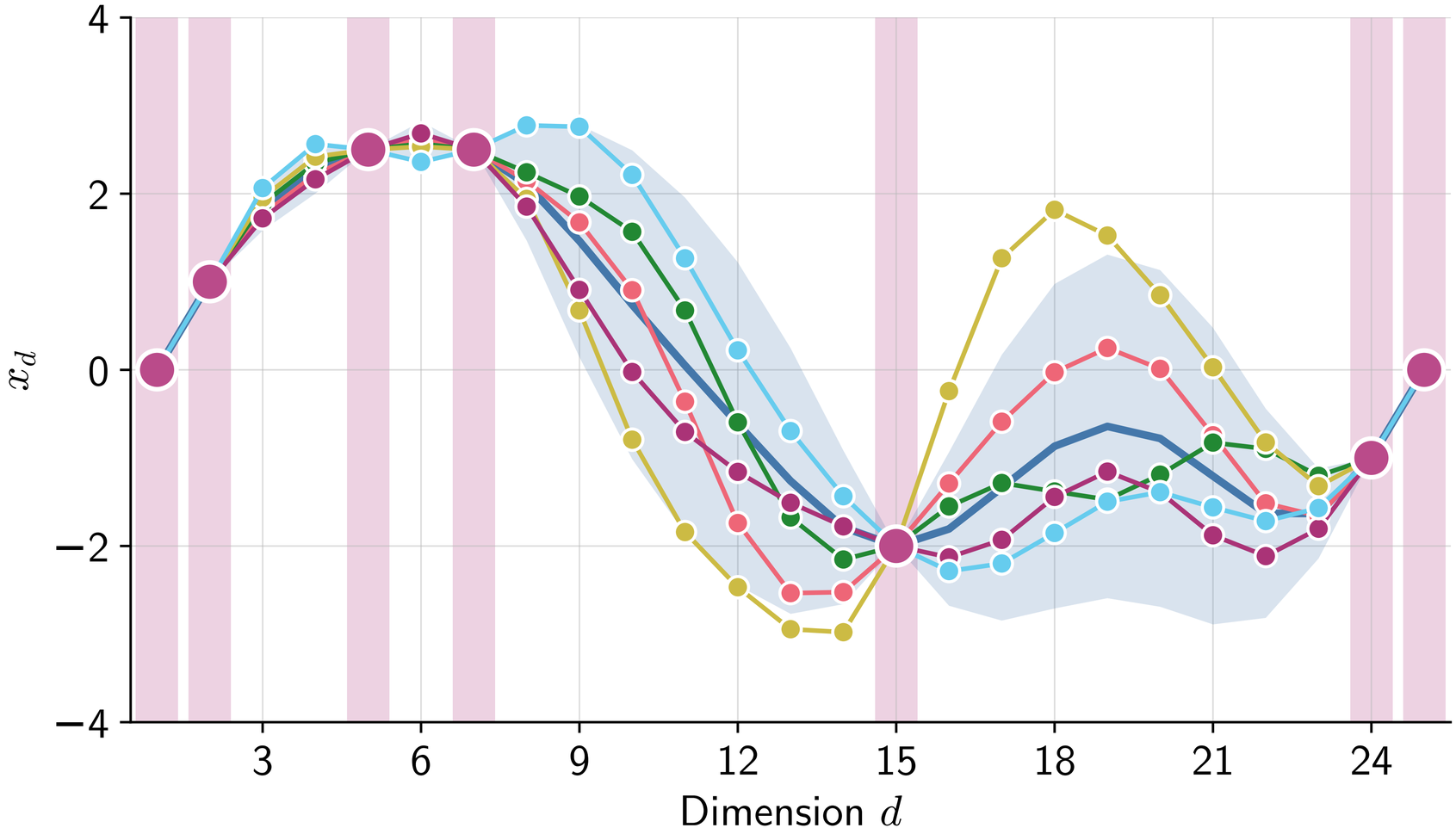
Conditioning a multivariate Gaussian



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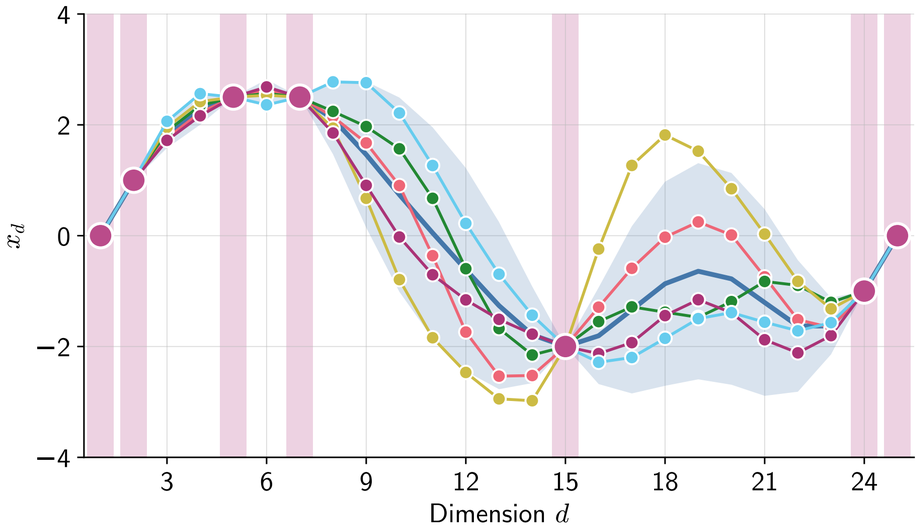


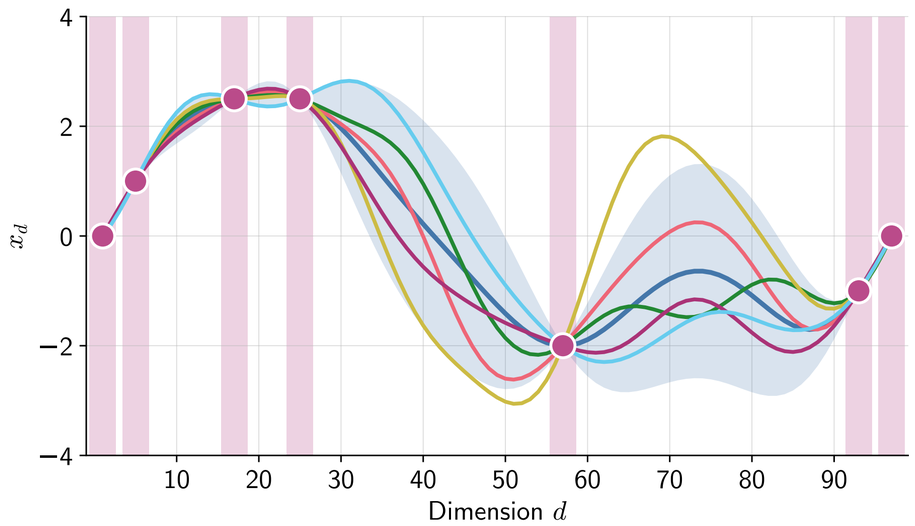
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- Let's now consider the case of $D = 100$

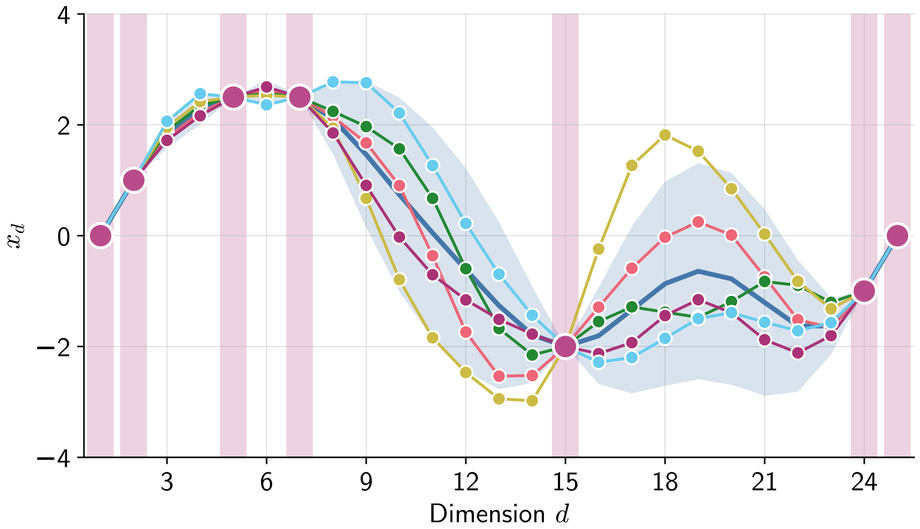


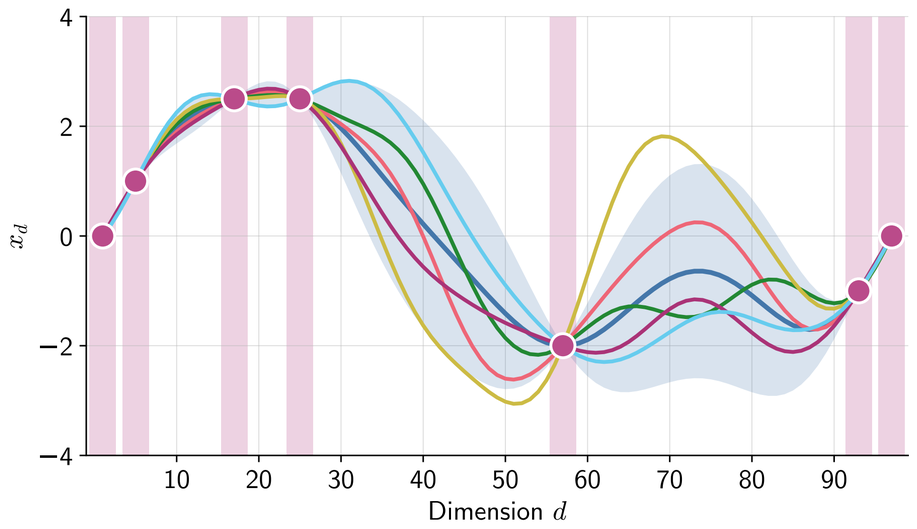


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- *Informally:* Infinitely long vectors can be seen as functions.

- By computing conditional Gaussian distributions, we can do non-linear regression!

Bayesian linear regression

- **Recall:** The linear regression model: $f(\mathbf{x}; \mathbf{w}) = \mathbf{x}^T\mathbf{w}$
- Let's define a prior (regulariser) on $\mathbf{w}$: $p(\mathbf{w}) = \mathcal{N}(0, \sigma_w^2\mathbf{I})$
- Now, define the likelihood as follows: $p(y \mid \mathbf{x}, \mathbf{w}) = \mathcal{N}(f(\mathbf{x}; \mathbf{w}), \sigma^2)$
- The joint is: $\begin{bmatrix}\mathbf{w}\\ \mathbf{y}\end{bmatrix} \sim p(\mathbf{w}, \mathbf{y} \mid \mathbf{X}) = \mathcal{N}\left(\mathbf{0}, \begin{bmatrix}\sigma_w^2\mathbf{I} & \sigma_w^2\mathbf{X}^T\\ \sigma_w^2\mathbf{X} & \sigma_w^2\mathbf{X}\mathbf{X}^T + \sigma^2\mathbf{I}\end{bmatrix}\right)$
- **Goal:** We want: $p(\mathbf{w} \mid \mathbf{X}, \mathbf{y})$ to perform predictions using

$$p(y_\star \mid \mathbf{x}_\star, \mathbf{X}, \mathbf{y}) = \int_{\mathbf{w}} p(y_\star \mid \mathbf{x}_\star, \mathbf{w})\, p(\mathbf{w} \mid \mathbf{X}, \mathbf{y})\, d\mathbf{w}$$

Recall: Conditioning a multivariate Gaussian

$$p(y \mid x) = \frac{p(x, y)}{p(x)}$$

- Let $\mathbf{x}_1$ and $\mathbf{x}_2$ be jointly Gaussian distributed:

$$p(\mathbf{x}_1, \mathbf{x}_2) = \mathcal{N}\left(\begin{bmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \end{bmatrix} \mid \begin{bmatrix} \mathbf{m}_1 \\ \mathbf{m}_2 \end{bmatrix}, \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix} \right)$$

- and

$$p(\mathbf{x}_2) = \int_{\mathbf{x}_1} p(\mathbf{x}_1, \mathbf{x}_2)\, d\mathbf{x}_1 = \mathcal{N}\left(\mathbf{x}_2 \mid \mathbf{m}_2, \Sigma_{22}\right)$$

- then

$$p(\mathbf{x}_1 \mid \mathbf{x}_2) = \mathcal{N}\left(\mathbf{x}_1 \mid \Sigma_{12}\Sigma_{22}^{-1}\left[\mathbf{x}_2 - \mathbf{m}_2\right] + \mathbf{m}_1, \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}\right)$$

Bayesian linear regression

- **Bayesian linear regression:** $\begin{bmatrix}\mathbf{w}\\ \mathbf{y}\end{bmatrix} \sim p(\mathbf{w}, \mathbf{y} \mid \mathbf{X}) = \mathcal{N}\left(\mathbf{0}, \begin{bmatrix}\sigma_w^2\mathbf{I} & \sigma_w^2\mathbf{X}^T \\ \sigma_w^2\mathbf{X} & \sigma_w^2\mathbf{X}\mathbf{X}^T + \sigma^2\mathbf{I}\end{bmatrix}\right)$
- Given: $p(\mathbf{x}_1, \mathbf{x}_2) = \mathcal{N}\left(\begin{bmatrix}\mathbf{x}_1\\ \mathbf{x}_2\end{bmatrix} \mid \begin{bmatrix}\mathbf{m}_1\\ \mathbf{m}_2\end{bmatrix}, \begin{bmatrix}\Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22}\end{bmatrix}\right)$
- We have: $p(\mathbf{x}_1 \mid \mathbf{x}_2) = \mathcal{N}\left(\mathbf{x}_1 \mid \Sigma_{12}\Sigma_{22}^{-1}\left[\mathbf{x}_2 - \mathbf{m}_2\right] + \mathbf{m}_1, \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}\right)$
- Hence, $p(\mathbf{w} \mid \mathbf{X}, \mathbf{y}) = \mathcal{N}\left(\mathbf{w} \mid \sigma_w^2\mathbf{X}^T\right.$

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Bayesian linear regression in feature spaces with map ϕ

- The linear regression model with: $f(\mathbf{x}; \mathbf{w}) = \phi(\mathbf{x})^T \mathbf{w}$
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$$p(\mathbf{w} \mid \mathbf{X}, \mathbf{y}) = \mathcal{N}\left(\mathbf{w} \mid \sigma_w^2 \mathbf{\Phi}^T \left(\sigma_w^2 \mathbf{\Phi}\mathbf{\Phi}^T + \sigma^2 \mathbf{I}\right)^{-1} \mathbf{y},\ \sigma_w^2 \mathbf{I} - \sigma_w^2 \mathbf{\Phi}^T \left(\sigma_w^2 \mathbf{\Phi}\mathbf{\Phi}^T + \sigma^2 \mathbf{I}\right)^{-1} \sigma_w^2 \mathbf{\Phi}\right)$$

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$$p(\mathbf{w} \mid \mathbf{X},\mathbf{y}) = \mathcal{N}\left(\mathbf{w} \mid \sigma_w^2\mathbf{\Phi}^T\left(\mathbf{K}+\sigma^2\mathbf{I}\right)^{-1}\mathbf{y},\ \sigma_w^2\mathbf{I} - \sigma_w^2\mathbf{\Phi}^T\left(\mathbf{K}+\sigma^2\mathbf{I}\right)^{-1}\sigma_w^2\mathbf{\Phi}\right)$$

- **Goal:** We want to perform predictions using $p(y_\star \mid \mathbf{x}_\star,\mathbf{X},\mathbf{y}) =$

$$\mathcal{N}\left(y_\star \mid \mathbf{k}_\star{}^T\left(\mathbf{K}+\sigma^2\mathbf{I}\right)^{-1}\mathbf{y},\ k(\mathbf{x}_\star,\mathbf{x}_\star) - \mathbf{k}_\star{}^T\left(\mathbf{K}+\sigma^2\mathbf{I}\right)^{-1}\mathbf{k}_\star + \sigma^2\right)$$

$$k(\mathbf{x},\mathbf{x}') = \sigma_w^2\,\phi(\mathbf{x})^T\phi(\mathbf{x}')$$
$$\mathbf{K} = \sigma_w^2\mathbf{\Phi}\mathbf{\Phi}^T$$
$$\mathbf{k}_\star = \sigma_w^2\mathbf{\Phi}\phi(\mathbf{x}_\star)$$

Gaussian process

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$$\mathbb{E}_{f\sim\mathrm{GP}}\left[f(\mathbf{x})\right] = m(\mathbf{x}) \qquad \mathbf{cov}\left[f(\mathbf{x}), f(\mathbf{x}')\right] = \mathbb{E}_{f\sim\mathrm{GP}}\left[(f(\mathbf{x}) - m(\mathbf{x}))(f(\mathbf{x}') - m(\mathbf{x}'))\right] = k(\mathbf{x}, \mathbf{x}')$$

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mean

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- Hence, $f(\mathbf{x})$ and $f(\mathbf{x}')$ are jointly Gaussian distributed with **covariance** $k(\mathbf{x}, \mathbf{x}')$

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- **Assumptions.** We will make the following assumptions (w.l.o.g.):
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  - $\mathbf{k}_\star = [k(\mathbf{x}_\star, \mathbf{x}_1), \ldots, k(\mathbf{x}_\star, \mathbf{x}_N)]^T$
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- Let $\mathbf{f} = [f(\mathbf{x}_1), \ldots, f(\mathbf{x}_N)]^T$, for any finite data set, $\mathbf{x}_1, \ldots, \mathbf{x}_N$ we have:

$$p(\mathbf{f}) = \mathcal{N}(\mathbf{f} \mid \mathbf{0}, \mathbf{K})$$

Gaussian process regression

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- Let $\mathbf{f} = [f(\mathbf{x}_1), \ldots, f(\mathbf{x}_N)]^T$ we write for the likelihood model:

$$p(\mathbf{y} \mid \mathbf{f}) = \mathcal{N}(\mathbf{y} \mid \mathbf{f}, \sigma^2 \mathbf{I})$$

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- Question: What is $f(\mathbf{x})$?
- We can place a **prior** on the function f: $p(\mathbf{f}) = \mathcal{N}(\mathbf{f} \mid \mathbf{0}, \mathbf{K})$
- Then, the joint for a new input $\mathbf{x}_\star$ is:

$$p(\mathbf{f}, f_\star) = \mathcal{N}\left(\begin{bmatrix} \mathbf{f} \\ f_\star \end{bmatrix} \mid \mathbf{0}, \begin{bmatrix} \mathbf{K} & \mathbf{k}_\star \\ \mathbf{k}_\star^T & k(\mathbf{x}_\star, \mathbf{x}_\star) \end{bmatrix} \right)$$

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- **Recall:** $p(\mathbf{x}_1 \mid \mathbf{x}_2) = \mathcal{N}\left(\mathbf{x}_1 \mid \Sigma_{12}\Sigma_{22}^{-1}\left[\mathbf{x}_2 - \mathbf{m}_2\right] + \mathbf{m}_1, \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}\right)$

  Different ordering!
- Hence, we have: $p(f_\star \mid \mathbf{f}) = \mathcal{N}\left(f_\star \mid \mathbf{k}_\star^T\right.$

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- Then, the joint for a new input $\mathbf{x}_\star$ is: $$p(\mathbf{f}, f_\star) = \mathcal{N}\left(\begin{bmatrix}\mathbf{f} \\ f_\star\end{bmatrix} \mid \mathbf{0}, \begin{bmatrix}\mathbf{K} & \mathbf{k}_\star \\ \mathbf{k}_\star^T & k(\mathbf{x}_\star, \mathbf{x}_\star)\end{bmatrix}\right)$$
- **Recall:** $p(\mathbf{x}_1 \mid \mathbf{x}_2) = \mathcal{N}\left(\mathbf{x}_1 \mid \Sigma_{12}\Sigma_{22}^{-1}\left[\mathbf{x}_2 - \mathbf{m}_2\right] + \mathbf{m}_1, \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}\right)$
- Hence, we have: $p(f_\star \mid \mathbf{f}) = \mathcal{N}\left(f_\star \mid \mathbf{k}_\star^T\right.$

Gaussian process regression

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- Hence, we have: $p(f_\star \mid \mathbf{f}) = \mathcal{N}\left(f_\star \mid \mathbf{k}_\star^T\mathbf{K}^{-1}\right.$

Gaussian process regression

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What if the data has noise?

Gaussian process regression - putting everything together

- **Prior** on function f: $p(\mathbf{f}) = \mathcal{N}(\mathbf{f} \mid \mathbf{0}, \mathbf{K})$
- Joint for **new input** $\mathbf{x}_\star$: $p(\mathbf{f}, f_\star) = \mathcal{N}\left(\begin{bmatrix} \mathbf{f} \\ f_\star \end{bmatrix} \mid \mathbf{0}, \begin{bmatrix} \mathbf{K} & \mathbf{k}_\star \\ \mathbf{k}_\star^T & k(\mathbf{x}_\star, \mathbf{x}_\star) \end{bmatrix} \right)$

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- **Likelihood**: $p(\mathbf{y} \mid \mathbf{f}) = \mathcal{N}(\mathbf{y} \mid \mathbf{f}, \sigma^2 \mathbf{I})$ (Noise model)

Gaussian process regression - putting everything together

- **Prior** on function f: $p(\mathbf{f}) = \mathcal{N}(\mathbf{f} \mid \mathbf{0}, \mathbf{K})$
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- **Likelihood**: $p(\mathbf{y} \mid \mathbf{f}) = \mathcal{N}(\mathbf{y} \mid \mathbf{f}, \sigma^2\mathbf{I})$
- Joint model: $p(\mathbf{y}, f_\star, \mathbf{f}) = p(\mathbf{y} \mid \mathbf{f})\, p(f_\star, \mathbf{f})$

How do we make predictions?

Gaussian process regression - putting everything together

- To make predictions, we want to compute $p(f_\star \mid \mathbf{y})$ and marginalise out (remove) the dependence on $\mathbf{f}$.

Gaussian process regression - putting everything together

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Gaussian process regression - putting everything together

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- **Predictions** with Gaussian process regression are again conditioning:

$$p(f_\star \mid \mathbf{y}) = \mathcal{N}\left(f_\star \mid \mathbf{k}_\star^T \left(\mathbf{K} + \sigma^2 \mathbf{I} \right)^{-1} \mathbf{y},\, k(\mathbf{x}_\star, \mathbf{x}_\star) - \mathbf{k}_\star^T \left(\mathbf{K} + \sigma^2 \mathbf{I} \right)^{-1} \mathbf{k}_\star \right)$$

Gaussian process regression - putting everything together

- Predictions **without** noise model:

$$p(f_\star \mid \mathbf{f}) = \mathcal{N}\left(f_\star \mid \mathbf{k}_\star^T \mathbf{K}^{-1} \mathbf{f},\, k(\mathbf{x}_\star, \mathbf{x}_\star) - \mathbf{k}_\star^T \mathbf{K}^{-1} \mathbf{k}_\star\right)$$

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Gaussian process regression - putting everything together

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- Predictions **with** noise model:

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- BLR with kernel $k(\mathbf{x}, \mathbf{x}') = \sigma_w^2 \boldsymbol{\phi}(\mathbf{x})^T \boldsymbol{\phi}(\mathbf{x}')$:

$$p(f_\star \mid \mathbf{x}_\star, \mathbf{X}, \mathbf{y}) = \mathcal{N}\left(f_\star \mid \mathbf{k}_\star^T \left(\mathbf{K} + \sigma^2 \mathbf{I}\right)^{-1} \mathbf{y},\, k(\mathbf{x}_\star, \mathbf{x}_\star) - \mathbf{k}_\star^T \left(\mathbf{K} + \sigma^2 \mathbf{I}\right)^{-1} \mathbf{k}_\star\right)$$

Gaussian process regression - putting everything together

- Predictions **without** noise model:

$$p(f_\star \mid \mathbf{f}) = \mathcal{N}\left(f_\star \mid \mathbf{k}_\star^T \mathbf{K}^{-1} \mathbf{f},\, k(\mathbf{x}_\star, \mathbf{x}_\star) - \mathbf{k}_\star^T \mathbf{K}^{-1} \mathbf{k}_\star\right)$$

- Predictions **with** noise model:

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- BLR with kernel $k(\mathbf{x}, \mathbf{x}') = \sigma_w^2 \boldsymbol{\phi}(\mathbf{x})^T \boldsymbol{\phi}(\mathbf{x}')$:

Identical

$$p(f_\star \mid \mathbf{x}_\star, \mathbf{X}, \mathbf{y}) = \mathcal{N}\left(f_\star \mid \mathbf{k}_\star^T \left(\mathbf{K} + \sigma^2 \mathbf{I}\right)^{-1} \mathbf{y},\, k(\mathbf{x}_\star, \mathbf{x}_\star) - \mathbf{k}_\star^T \left(\mathbf{K} + \sigma^2 \mathbf{I}\right)^{-1} \mathbf{k}_\star\right)$$

Implementing Gaussian process regression

Gaussian process regression

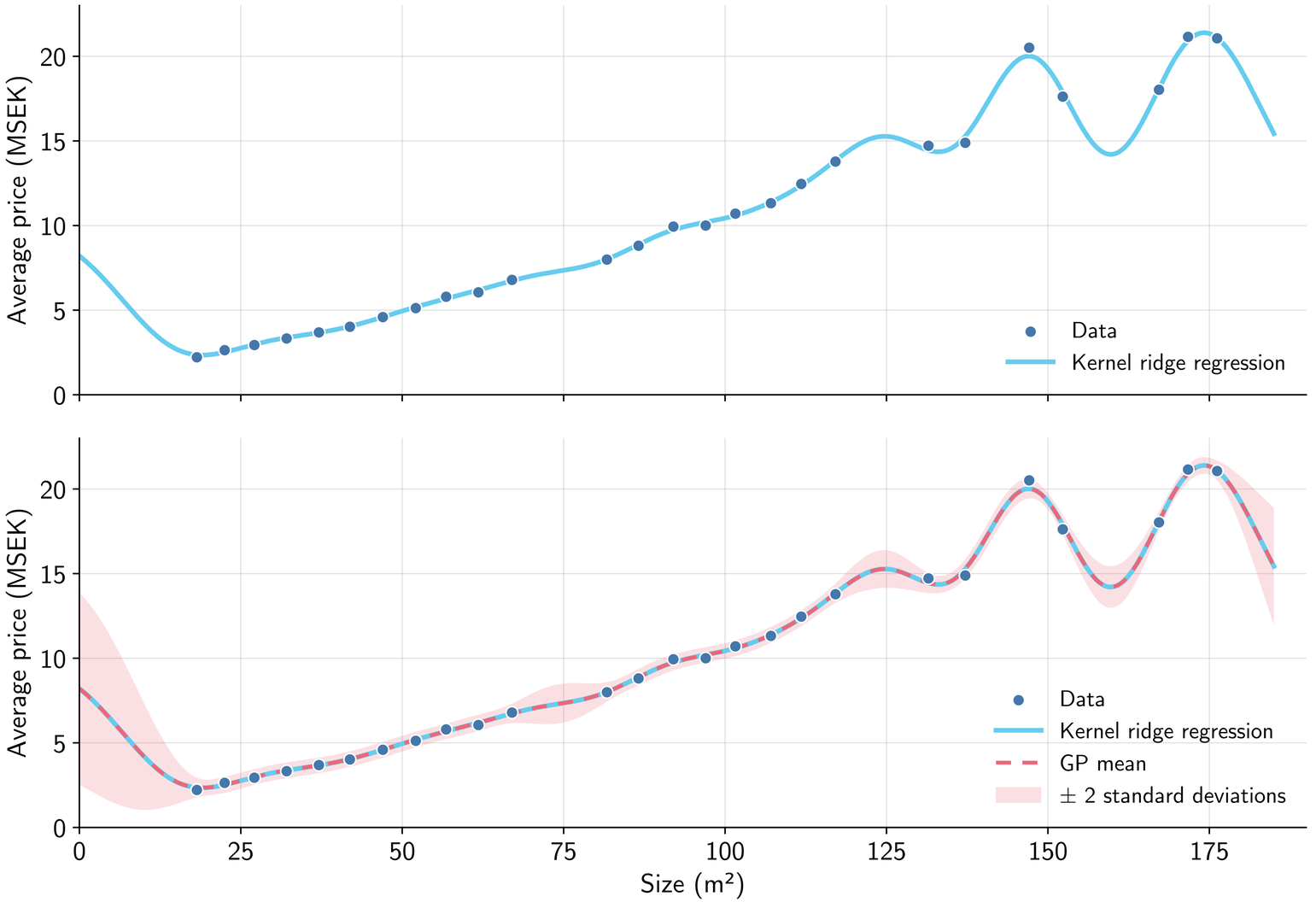
```python
# reuse rbf_kernel and gram_matrix from the last lecture
sigma2 = 0.1                              # noise variance σ²

K = gram_matrix(X, X)                     # K_ff
K_s = gram_matrix(X_star, X)              # K_f*f
K_ss = gram_matrix(X_star, X_star)        # K_f*f*
A = K + sigma2 * np.eye(len(X))           # K_ff + σ²I

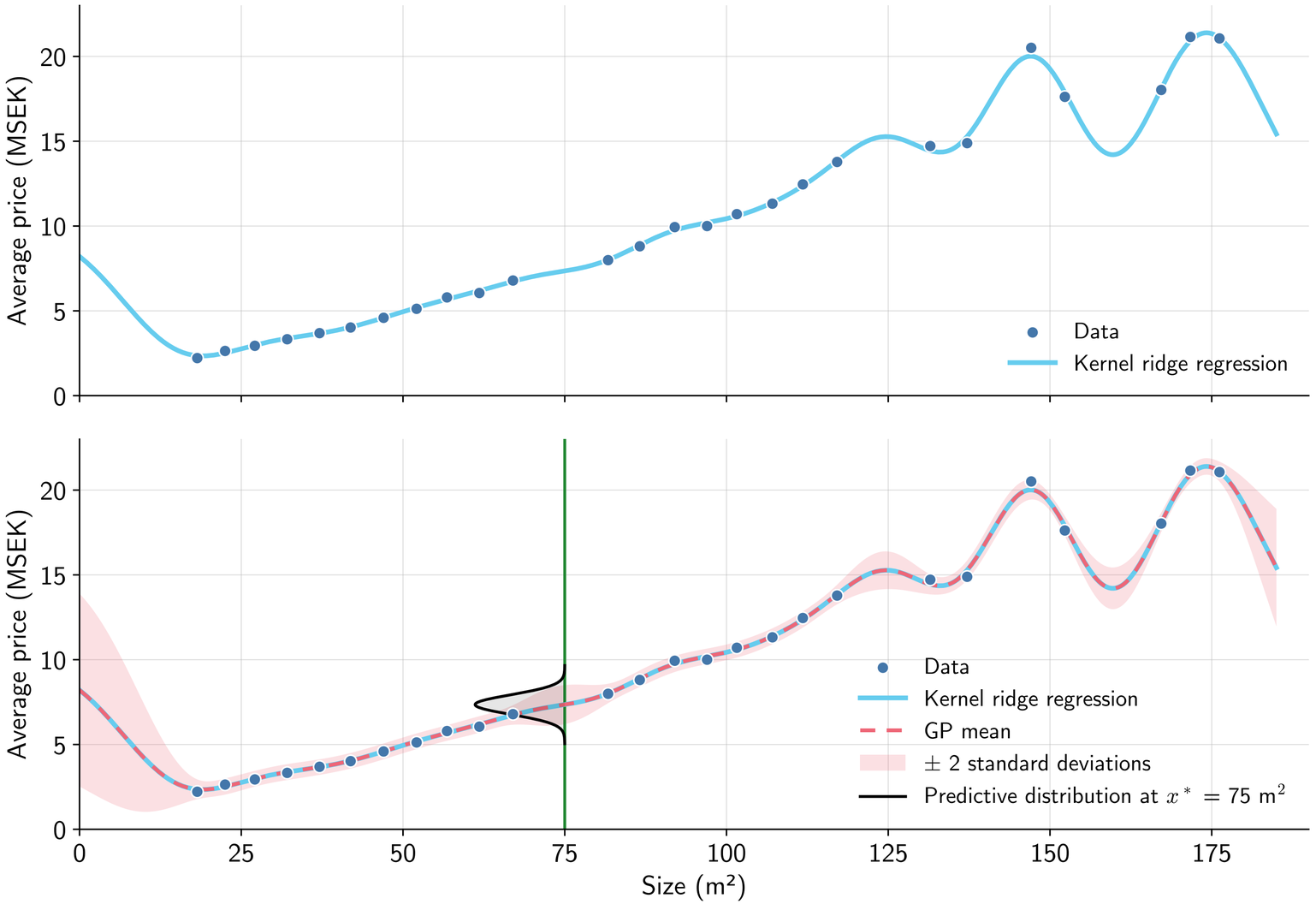
mean = K_s @ np.linalg.solve(A, y)
cov = K_ss - K_s @ np.linalg.solve(A, K_s.T)
std = np.sqrt(np.diag(cov))
```

https://github.com/traml-lab/id2214-lectures/blob/main/lecture6-gaussian-process.py

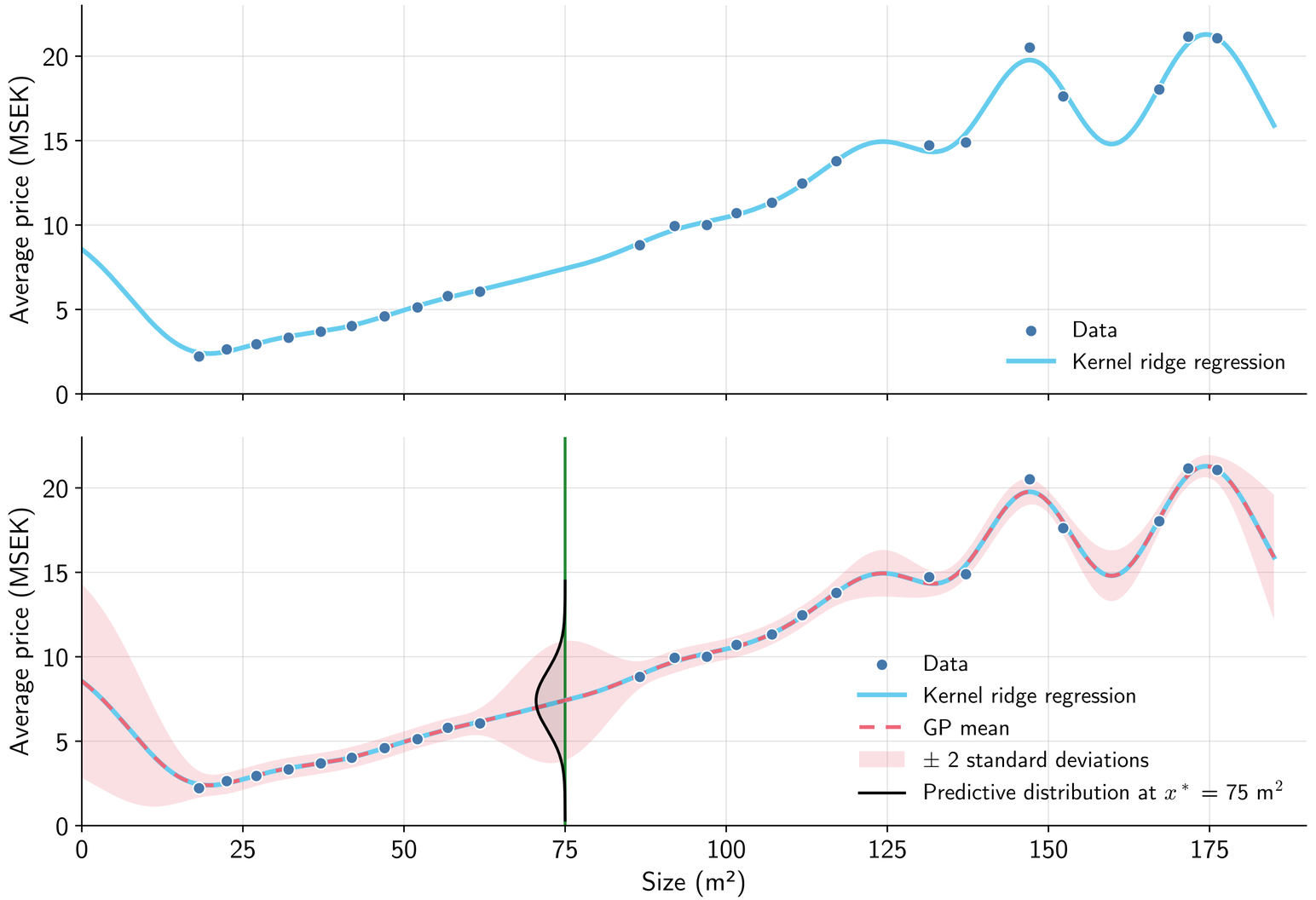
Gaussian process regression



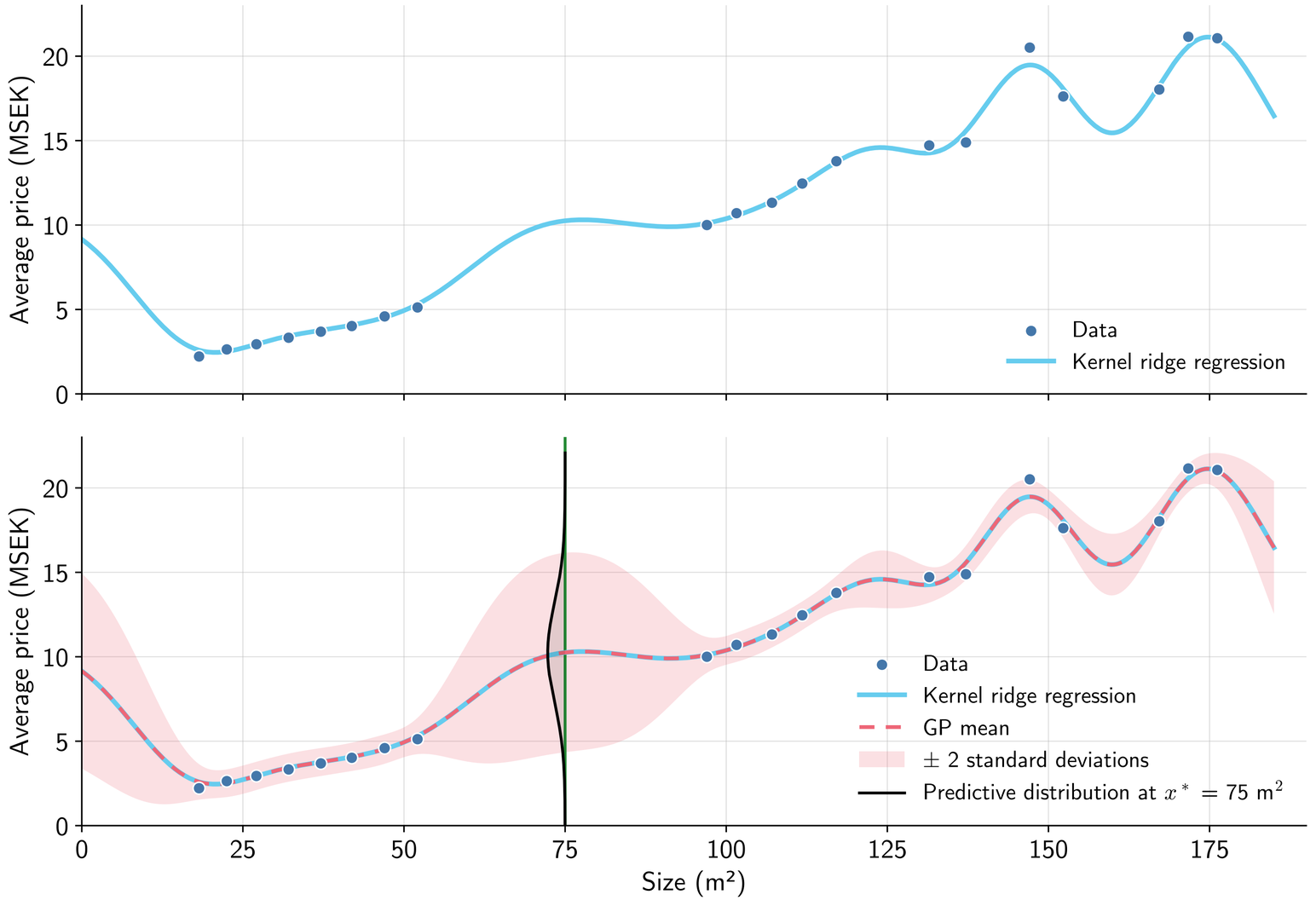
Gaussian process regression



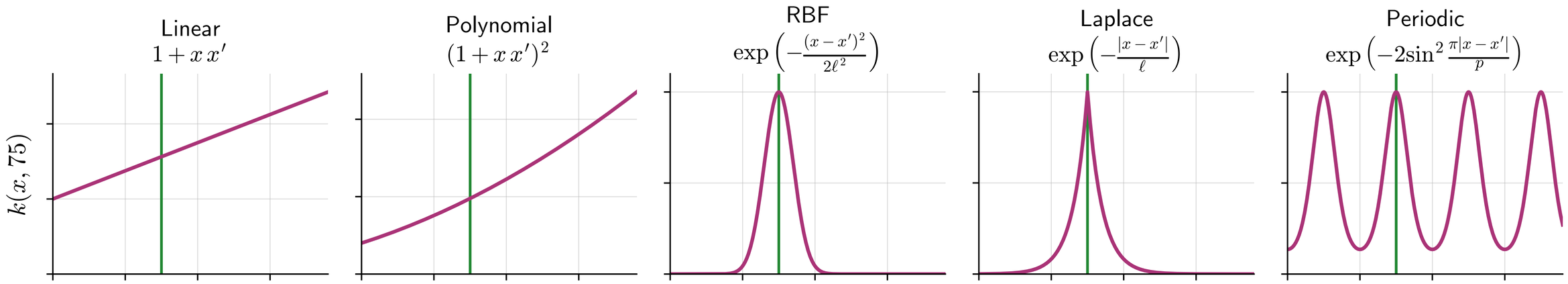
Gaussian process regression



Gaussian process regression



Kernel functions

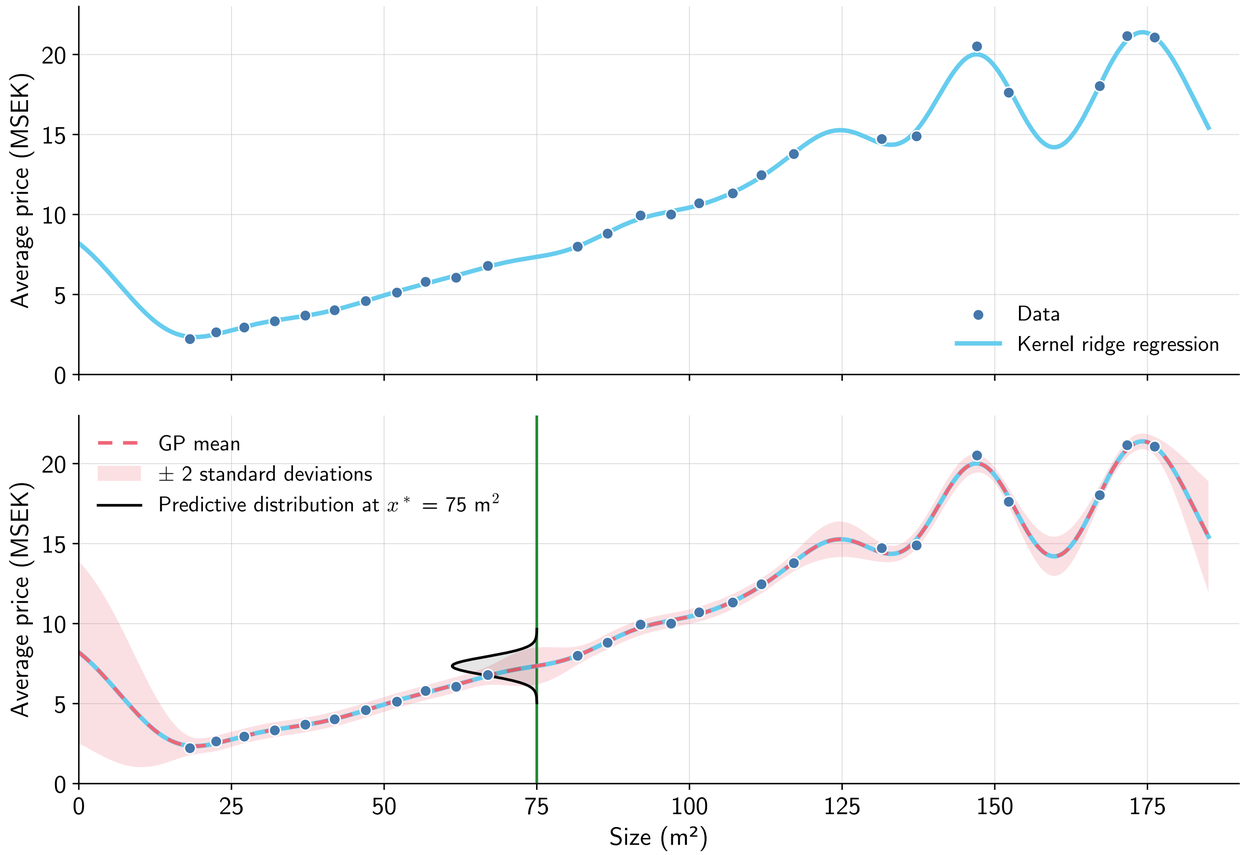


- In Gaussian processes, the kernel function is also a design choice!

$$p(f_\star \mid \mathbf{y}) = \mathcal{N}\left(f_\star \mid \mathbf{k}_\star^T \left(\mathbf{K} + \sigma^2\mathbf{I}\right)^{-1} \mathbf{y},\, k(\mathbf{x}_\star, \mathbf{x}_\star) - \mathbf{k}_\star^T \left(\mathbf{K} + \sigma^2\mathbf{I}\right)^{-1} \mathbf{k}_\star\right)$$

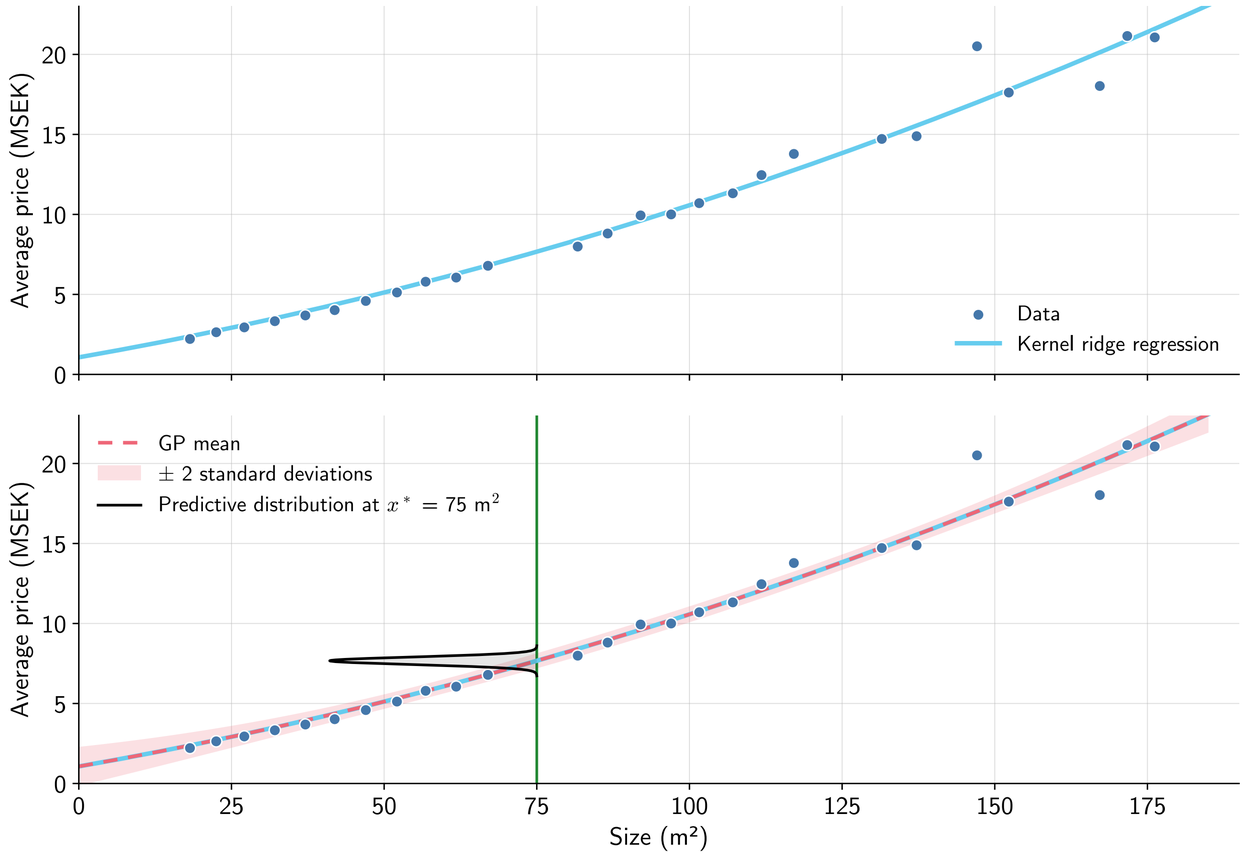
Gaussian process regression - kernel functions

$$k(\mathbf{x}, \mathbf{x}') = \sigma_f^2 \exp\left(-\frac{\|\mathbf{x} - \mathbf{x}'\|^2}{2\ell^2}\right)$$



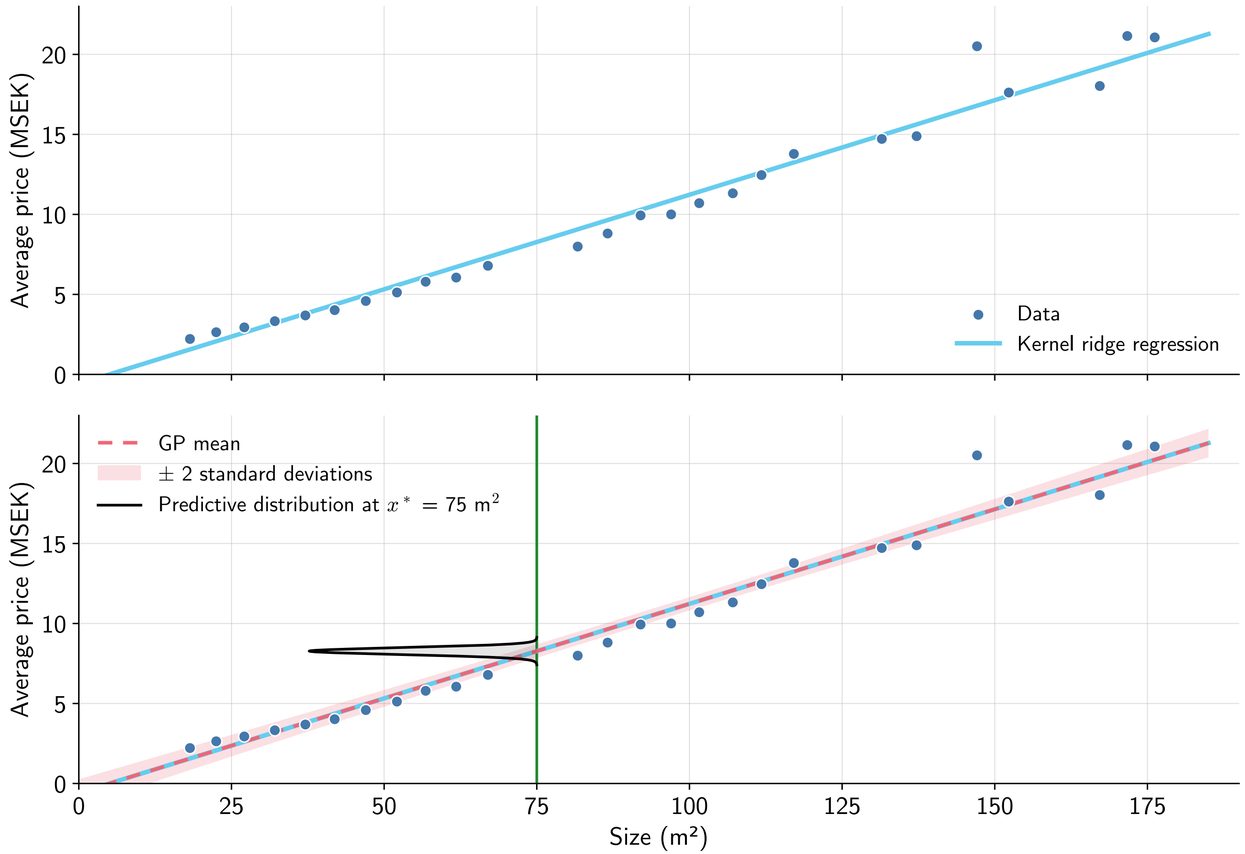
Gaussian process regression - kernel functions

$$k(\mathbf{x}, \mathbf{x}') = \sigma_f^2 \left(1 + \mathbf{x}^T \mathbf{x}'\right)^2$$



Gaussian process regression - kernel functions

$$k(\mathbf{x}, \mathbf{x}') = \sigma_f^2 \left(1 + \mathbf{x}^T \mathbf{x}'\right)$$



Gaussian process regression

- A Gaussian process is a collection of random variables indexed over space (or time), any finite subset of which has a joint Gaussian distribution.

- A Gaussian process (GP) can be considered to be a Bayesian prior distribution over functions: $f(\mathbf{x}) \sim \mathrm{GP}(m(\mathbf{x}), k(\mathbf{x}, \mathbf{x}'))$

- A GP is fully specified by its mean $m(\,\cdot\,)$ and its covariance/kernel function $k(\,\cdot\,,\cdot\,)$, but we typically assume a zero-mean $m(\mathbf{x}) = \mathbf{0}$ function for simplicity

- Predictions are conditional probabilities of multivariate Gaussians:

$$p(f_\star \mid \mathbf{y}) = \mathcal{N}\left(f_\star \mid \mathbf{k}_\star^T \left(\mathbf{K} + \sigma^2\mathbf{I}\right)^{-1} \mathbf{y},\, k(\mathbf{x}_\star, \mathbf{x}_\star) - \mathbf{k}_\star^T \left(\mathbf{K} + \sigma^2\mathbf{I}\right)^{-1} \mathbf{k}_\star\right)$$

The End

